Lesson 1 — Extra Practice

Review: The Pythagorean Theorem

Part A — Exact Answers: Hypotenuse and Missing Leg

For questions 1–4, give the **exact** answer as a simplified radical (as in Unit 1), and then a decimal rounded to **one decimal place**. If the answer is a whole number, say so.

1. Find the length of the hypotenuse of each right triangle.

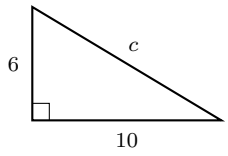
(a) Legs 5 and 7.

(b) Legs 6 and 9.

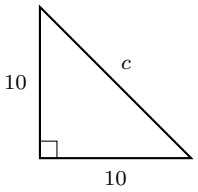
(c) Legs 3 and 11.

2. Find the length of the hypotenuse c in each right triangle below.

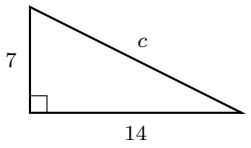
(a)



(b)



(c)



3. Find the length of the missing leg of each right triangle.

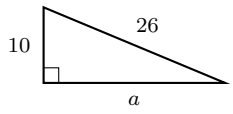
(a) Hypotenuse 15, one leg 9.

(b) Hypotenuse 20, one leg 8.

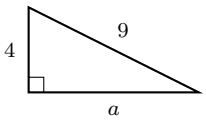
(c) Hypotenuse 14, one leg 5.

4. Find the length of the missing leg a in each right triangle below.

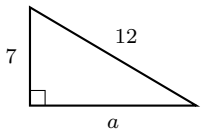
(a)



(b)



(c)



Part B — The Converse and Pythagorean Triples

The Pythagorean theorem also runs *backwards*. If the three side lengths of a triangle satisfy $a^2 + b^2 = c^2$, where c is the longest side, then the triangle **must** be a right triangle. This is called the **converse** of the theorem, and it is how you test a triangle you cannot measure with a protractor.

5. Use the converse to decide whether each set of side lengths forms a right triangle. Identify the longest side first, then show both sides of the equation and state your verdict.

(a) 20, 21, 29

(b) 8, 11, 14

(c) 16, 12, 20

6. Three famous **Pythagorean triples** are $3, 4, 5$ and $5, 12, 13$ and $8, 15, 17$. Every whole-number multiple of a triple is also a triple — for example, $3, 4, 5$ doubled gives $6, 8, 10$. For each set below, state whether it is a multiple of one of the three triples named above; if it is, say which triple and what the multiplier is.

(a) 15, 20, 25 (b) 24, 45, 51 (c) 25, 60, 65 (d) 9, 40, 41

7. Each triangle below is a scaled copy of one of the three triples in question 6. Find the missing side **without** squaring anything: spot the triple, find the multiplier, then scale the third member.

(a) Legs 21 and 28; find the hypotenuse.

(b) Legs 20 and 48; find the hypotenuse.

(c) Hypotenuse 85, one leg 40; find the other leg.

(d) Hypotenuse 39, one leg 15; find the other leg.

8. **Error analysis.** A right triangle has a hypotenuse of 17 cm and one leg of 15 cm. Dev wants the other leg, so he writes:

$$15^2 + 17^2 = c^2, \qquad 225 + 289 = 514, \qquad c = \sqrt{514} \approx 22.7 \text{ cm}.$$

(a) Explain what Dev did wrong. Which letter in $a^2+b^2 = c^2$ did he put in the wrong place?

(b) Without calculating, explain how you can tell his answer of 22.7 cm is impossible. Then find the correct length of the missing leg.

Part C — Two-Stage and Three-Dimensional Problems

Each question below needs the theorem used **more than once**, or needs a second step after the theorem. Label the side you find in stage one — you will need it in stage two.

9. A 13 ft ladder leans against a vertical wall with its base 5 ft from the wall.

(a) How high up the wall does the ladder reach?

(b) The base is then pulled out until it is 9 ft from the wall. How high does the ladder reach now? Give the exact height as a simplified radical and a decimal to one place.

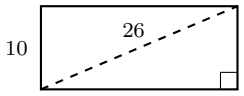
(c) By how much did the top of the ladder drop?

10. A rectangular window has a diagonal of 26 cm and a shorter side of 10 cm.

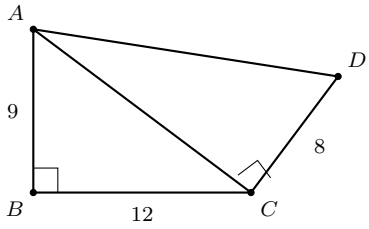
(a) Find the length of the longer side.

(b) Find the perimeter of the window.

(c) Find the area of the window.



11. In the figure, $\triangle ABC$ has a right angle at B, and $\triangle ACD$ has a right angle at C. The two triangles share the side $\overline{AC}$.

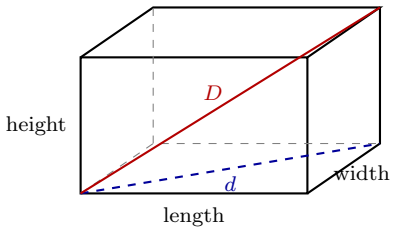


(a) Use $\triangle ABC$ to find the length of $\overline{AC}$.

(b) Use your answer from (a), together with $\triangle ACD$, to find the length of $\overline{AD}$.

(c) Both answers turned out to be whole numbers. Name the Pythagorean triple used in each stage.

12. **The space diagonal of a box (3-D stretch).** The longest straight rod that fits inside a rectangular box runs from one bottom corner to the opposite top corner. Finding it takes the theorem *twice*: first across the **base** of the box to get the base diagonal d, then up the **height** using d as a leg.



For each box, find the base diagonal d first, then the space diagonal D. Give exact simplified radicals and decimals to one place.

(a) A box 6 cm long, 8 cm wide and 10 cm tall.

(b) A box 4 in long, 4 in wide and 7 in tall. *Note:* the base diagonal here is *not* a whole number, but the space diagonal is — keep d^2 rather than d in the second stage.

13. **Reasoning.** Can a right triangle have two sides of equal length?
(a) Suppose the two *legs* are both 6 cm. Find the hypotenuse exactly and as a decimal to one place.
(b) Suppose instead a leg equals the hypotenuse, both 6 cm. What would the other leg have to be? Explain why this is impossible.

14. **Reasoning.** A right triangle has legs a and b and hypotenuse c.
(a) If both legs are doubled, what happens to the hypotenuse? Test your answer with the legs 3 and 4.
(b) Explain why c must always be less than $a + b$, no matter what the legs are.

Part D — Applications

Draw and label a right triangle for each problem before you calculate. Round every final answer to **one decimal place** unless it is exact.

15. A wheelchair ramp rises 2 m vertically over a horizontal run of 7 m. How long is the sloped surface of the ramp?

16. A guy wire 20 m long runs from the top of a straight vertical radio mast to an anchor in the ground 9 m from the base of the mast. How tall is the mast?

17. A softball diamond is a square with sides of 27 m. The catcher at home plate throws straight across the diamond to second base. How far is that throw? Give the exact distance as a simplified radical and a decimal to one place.

18. Television screens are advertised by the length of their diagonal. A screen measures 32 inches across and 18 inches high. What size would this television be advertised as, to one decimal place?

19. **Stretch.** A shipping crate measures 3 ft by 4 ft by 12 ft.

(a) Find the diagonal of the 3 ft by 4 ft base.

(b) Find the space diagonal of the crate.

(c) A metal rod 13.5 ft long must be shipped. Will it fit inside the crate? Justify your answer.

Answer Key. 1. (a) $\sqrt{74} \approx 8.6$ (b) $\sqrt{117} = 3\sqrt{13} \approx 10.8$ (c) $\sqrt{130} \approx 11.4$. **2.** (a) $\sqrt{136} = 2\sqrt{34} \approx 11.7$ (b) $\sqrt{200} = 10\sqrt{2} \approx 14.1$ (c) $\sqrt{245} = 7\sqrt{5} \approx 15.7$. **3.** (a) $\sqrt{225-81} = \sqrt{144} = 12$ exactly (b) $\sqrt{400-64} = \sqrt{336} = 4\sqrt{21} \approx 18.3$ (c) $\sqrt{196-25} = \sqrt{171} = 3\sqrt{19} \approx 13.1$. **4.** (a) $\sqrt{676-100} = \sqrt{576} = 24$ exactly (b) $\sqrt{81-16} = \sqrt{65} \approx 8.1$ (c) $\sqrt{144-49} = \sqrt{95} \approx 9.7$. **5.** (a) longest 29: $20^2 + 21^2 = 400 + 441 = 841$ and $29^2 = 841$ — **right**. (b) longest 14: $8^2+11^2 = 64+121 = 185$ but $14^2 = 196$; $185 \neq 196$ — **not right**. (c) longest 20: $12^2+16^2 = 144+256 = 400$ and $20^2 = 400$ — **right**. **6.** (a) $3, 4, 5$ times 5 (b) $8, 15, 17$ times 3 (c) $5, 12, 13$ times 5 (d) not a multiple of any of the three — but $81+1600 = 1681 = 41^2$, so $9, 40, 41$ is a Pythagorean triple in its own right. **7.** (a) $3, 4, 5$ times 7, so $c = 35$ (b) $5, 12, 13$ times 4, so $c = 52$ (c) $8, 15, 17$ times 5, so the other leg is 75 (d) $5, 12, 13$ times 3, so the other leg is 36. **8.** (a) He used the hypotenuse 17 as a leg. The hypotenuse must be c, alone on its own side: the correct set-up is $15^2 + b^2 = 17^2$. (b) The hypotenuse is the longest side, so no other side can be longer than 17 cm — 22.7 cm is impossible. Correctly, $b = \sqrt{289-225} = \sqrt{64} = 8$ cm. **9.** (a) $\sqrt{169-25} = \sqrt{144} = 12$ ft. (b) $\sqrt{169-81} = \sqrt{88} = 2\sqrt{22} \approx 9.4$ ft. (c) $12 - 9.4 = 2.6$ ft lower (exactly $12 - 2\sqrt{22}$). **10.** (a) $\sqrt{676-100} = \sqrt{576} = 24$ cm. (b) $P = 2(24) + 2(10) = 68$ cm. (c) $A = 24 \times 10 = 240$ cm^2. **11.** (a) $AC = \sqrt{81+144} = \sqrt{225} = 15$. (b) $AD = \sqrt{15^2+8^2} = \sqrt{225+64} = \sqrt{289} = 17$. (c) Stage one is $9, 12, 15$ ($3, 4, 5$ times 3); stage two is $8, 15, 17$. **12.** (a) $d = \sqrt{36+64} = \sqrt{100} = 10$ cm; $D = \sqrt{100+100} = \sqrt{200} = 10\sqrt{2} \approx 14.1$ cm. (b) $d = \sqrt{16+16} = \sqrt{32} = 4\sqrt{2} \approx 5.7$ in, so $d^2 = 32$; $D = \sqrt{32+49} = \sqrt{81} = 9$ in exactly. **13.** (a) $c = \sqrt{36+36} = \sqrt{72} = 6\sqrt{2} \approx 8.5$ cm — yes, the two *legs* may be equal. (b) It would need $a^2 + 36 = 36$, so $a = 0$, which is not a triangle. The hypotenuse is always the longest side, so it can never equal a leg. **14.** (a) The hypotenuse doubles: $\sqrt{(2a)^2 + (2b)^2} = \sqrt{4(a^2+b^2)} = 2\sqrt{a^2+b^2} = 2c$. Check: legs $3, 4$ give $c = 5$; legs $6, 8$ give $c = 10$. (b) Squaring, $c^2 = a^2 + b^2$ while $(a+b)^2 = a^2 + 2ab + b^2$, which is larger by $2ab > 0$. (Geometrically, the straight path along the hypotenuse beats going around the two legs.) **15.** $\sqrt{4+49} = \sqrt{53} \approx 7.3$ m. **16.** $\sqrt{400-81} = \sqrt{319} \approx 17.9$ m. **17.** $\sqrt{27^2+27^2} = \sqrt{1458} = 27\sqrt{2} \approx 38.2$ m. **18.** $\sqrt{1024+324} = \sqrt{1348} = 2\sqrt{337} \approx 36.7$ inches, so it would be advertised as a 36.7-inch television. **19.** (a) $\sqrt{9+16} = \sqrt{25} = 5$ ft. (b) $\sqrt{25+144} = \sqrt{169} = 13$ ft. (c) No. The longest object that fits is the space diagonal, 13 ft, and $13.5 > 13$, so the rod is too long by 0.5 ft.

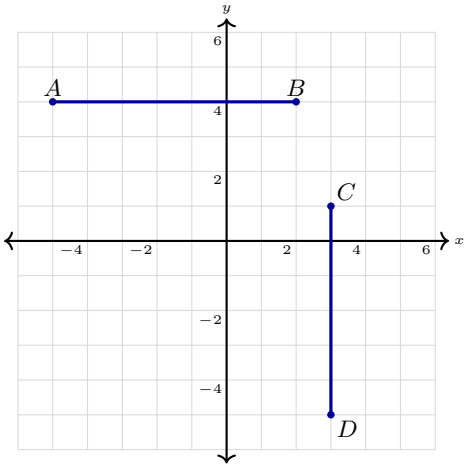
Lesson 2 — Extra Practice

Measuring the Length of a Line Segment

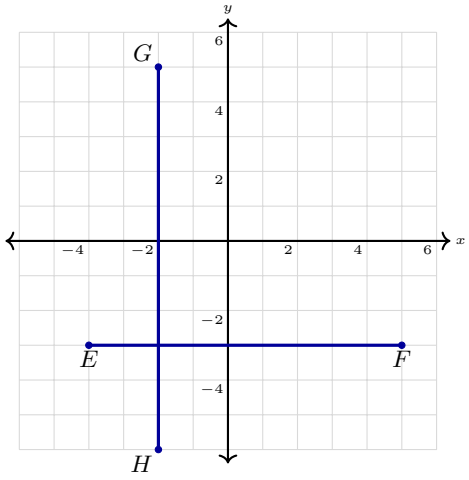
Part A — Counting Squares and Subtracting Coordinates

Remember the swap: a **horizontal** segment shares a y-coordinate and uses the x's, so length $= |x_2 - x_1|$; a **vertical** segment shares an x-coordinate and uses the y's, so length $= |y_2 - y_1|$.

1. Four segments are graphed below. For each one, state its endpoints, say whether it is horizontal or vertical, find its length by **counting squares**, and then confirm that length by **subtracting coordinates**.



(a) $\overline{AB}$ (b) $\overline{CD}$



(c) $\overline{EF}$ (d) $\overline{GH}$

2. Determine the length of each segment. Watch the negatives — show the subtraction.

(a) $A(-7, 3)$ to $B(5, 3)$ (b) $C(-2, -9)$ to $D(-2, -1)$ (c) $E(-11, -4)$ to $F(-3, -4)$ (d) $G(6, -13)$ to $H(6, 12)$

3. These points are too far apart to graph. Determine each length by subtracting.

(a) $J(-45, 20)$ to $K(68, 20)$ (b) $L(-17, -90)$ to $M(-17, 55)$ (c) $N(-126, -8)$ to $P(-39, -8)$ (d) $Q(14, -205)$ to $R(14, -68)$

4. The rule works for coordinates that are not whole numbers too. Determine each length.

(a) $S(2.5, -6)$ to $T(-8.5, -6)$ (b) $U(-3.75, 4)$ to $V(-3.75, 9.25)$ (c) $W\left(\frac{1}{2}, -3\right)$ to $X\left(-\frac{7}{2}, -3\right)$ (d) $Y\left(-6, \frac{5}{4}\right)$ to $Z\left(-6, -\frac{11}{4}\right)$

5. State whether each segment is horizontal or vertical, and write an expression for its length. Assume every letter stands for a real number.

(a) $A(p, q)$ to $B(p, q - 12)$

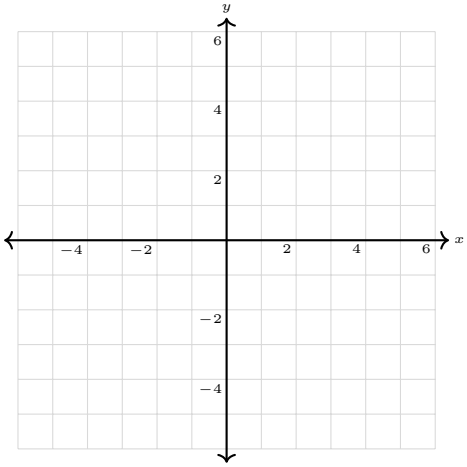
(b) $C(m + 9, n)$ to $D(m - 4, n)$

(c) $E(a, b)$ to $F(a, c)$, where $c > b$

(d) $G(r, s)$ to $H(t, s)$, where $r > t$

Part B — Perimeter and Figures on the Grid

6. Rectangle $ABCD$ has vertices $A(-5, -3)$, $B(4, -3)$, $C(4, 4)$ and $D(-5, 4)$.



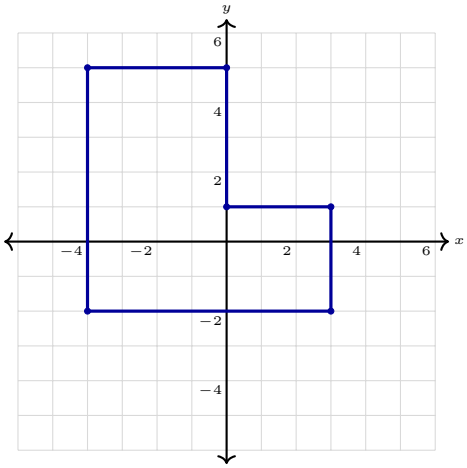
(a) Plot and label the four vertices, then join them in order.

(b) Find the length of each of the four sides by subtracting coordinates.

(c) Determine the perimeter of $ABCD$.

(d) Determine the area of $ABCD$.

7. An L-shaped room is drawn on a floor plan with corners at $(-4, -2)$, $(3, -2)$, $(3, 1)$, $(0, 1)$, $(0, 5)$ and $(-4, 5)$, joined in that order and then back to the start. Each grid unit represents one metre.

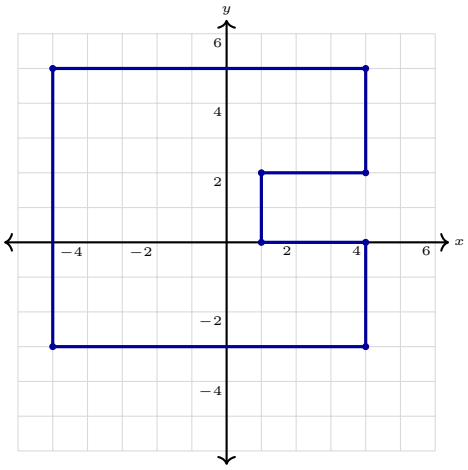


(a) Find the length of all six sides. Organise your work in a list so none is missed.

(b) Determine the perimeter of the room, in metres.

(c) Baseboard costs \$6.50 per metre. Find the cost of baseboard for the whole room.

8. A swimming pool deck is shaped like the figure below, with corners at $A(-5,-3)$, $B(4,-3)$, $C(4,0)$, $D(1,0)$, $E(1,2)$, $F(4,2)$, $G(4,5)$ and $H(-5,5)$, joined in order and back to A. Notice that the figure has a **notch** cut into its right side.



(a) Find the length of all eight sides.

(b) Determine the perimeter of the deck.

(c) The smallest rectangle that would enclose this figure runs from $x=-5$ to $x=4$ and from $y=-3$ to $y=5$. Find that rectangle's perimeter and compare it to your answer in (b). Explain why the notch makes the two answers differ.

9. Five segments are described below. Determine the length of each, then list the segments in order from **shortest to longest**, and state which is longest.

P_1: $(-8,6)$ to $(7,6)$
P_2: $(3,-9)$ to $(3,8)$
P_3: $(-12,-5)$ to $(2,-5)$
P_4: $(-6,11)$ to $(-6,-2)$
P_5: $(5,-4)$ to $(21,-4)$

Part C — Working Backwards

In these questions the *length* is given and an endpoint is missing. Because a segment can run in either direction from a known endpoint, expect **two** answers unless the question rules one out.

10. One endpoint and a length are given. Find **both** possible positions of the other endpoint.

(a) Horizontal, from $A(3,-2)$, length 7
(b) Vertical, from $B(-5,4)$, length 9
(c) Horizontal, from $C(-1,-6)$, length 12
(d) Vertical, from $D(7,2)$, length 15

11. Now a condition is added, so only **one** answer survives. Solve for the unknown coordinate and state the other endpoint.

(a) $F(4,-3)$ to $G(x,-3)$, with $FG=10$ and $x<4$.
(b) $H(-2,y)$ to $J(-2,5)$, with $HJ=11$ and $y>5$.
(c) $K(x,-8)$ to $L(-9,-8)$, with $KL=6$ and $x<-9$.

12. One side of a rectangle joins $A(-3, -4)$ to $B(9, -4)$, and the rectangle's perimeter is 40 units. Find the coordinates of the other two vertices. There are two possible rectangles — give both, and sketch a rough diagram of each.

13. *(Multiple Choice)* Vertical segment MN has endpoints $M(-4, 7)$ and $N(-4, y)$, and $MN = 15$ units. Which statement gives all possible values of y?
A. $y = 22$ only B. $y = -8$ only
C. $y = 22$ or $y = -8$ D. $y = 11$ or $y = -11$

Part D — Error Analysis, Reasoning, and a Stretch

14. **Error analysis.** Two students found the length of the segment with endpoints $R(-3, 7)$ and $S(9, 7)$.
(a) Maya wrote: "length $= -3 - 9 = -12$, so $RS = -12$ units." Identify her error, explain in one sentence why her answer cannot be right, and give the correct length.
(b) Owen wrote: "the y's are both 7, so $RS = 7 - 7 = 0$ units." Identify his error and explain which coordinates he should have subtracted, and why.

15. **Reasoning.** Explain, in your own words, what the absolute value bars in length $= |x_2 - x_1|$ are doing. In your explanation, say why subtracting in the opposite order is *not* an error, and why a length can never be negative.

16. **Reasoning.** Nina says: "If both endpoints have negative x-coordinates, the length must be negative too." Give a counterexample with your own coordinates, and explain the flaw in her thinking.

17. A rectangular hockey rink is laid out on a plan with corners at $(-40, -15)$, $(60, -15)$, $(60, 35)$ and $(-40, 35)$, in metres. Find the perimeter of the rink, and the total cost of boards around it at \$85 per metre.

18. A city block is bounded by streets running from $(0, 0)$ to $(240, 0)$, then to $(240, 180)$, then to $(0, 180)$, then back to the start, in metres. A jogger runs three full laps around the block. How far does she run, in kilometres?

19. **Stretch.** A rectangle has its sides parallel to the axes, a perimeter of 30 units, whole-number side lengths, and one vertex at the point $(2, 3)$.

(a) If the width is w and the height is h, explain why $w + h = 15$. Then list every ordered pair of whole numbers (w, h) that works ($(1, 14)$ and $(14, 1)$ count as two).

(b) Fix one such pair, say $w = 4$ and $h = 11$. The point $(2, 3)$ could be any one of the rectangle's four corners. Give the other three vertices for **each** of those four rectangles.

(c) Counting every width, every height, and every corner, how many different rectangles satisfy all the conditions? Explain your count.

Answer Key. 1. (a) $A(-5, 4)$, $B(2, 4)$; horizontal (the y's match); $|2 - (-5)| = 7$ units. (b) $C(3, 1)$, $D(3, -5)$; vertical; $|1 - (-5)| = 6$ units. (c) $E(-4, -3)$, $F(5, -3)$; horizontal; $|5 - (-4)| = 9$ units. (d) $G(-2, 5)$, $H(-2, -6)$; vertical; $|5 - (-6)| = 11$ units. **2.** (a) $|5 - (-7)| = 12$ (b) $|-1-(-9)| = 8$ (c) $|-3-(-11)| = 8$ (d) $|12 - (-13)| = 25$. **3.** (a) $|68 - (-45)| = 113$ (b) $|55 - (-90)| = 145$ (c) $|-39-(-126)| = 87$ (d) $|-68-(-205)| = 137$. **4.** (a) $|-8.5 - 2.5| = 11$ (b) $|9.25 - 4| = 5.25$ (c) $\left|-\frac{7}{2} - \frac{1}{2}\right| = 4$ (d) $\left|-\frac{11}{4} - \frac{5}{4}\right| = \frac{16}{4} = 4$. **5.** (a) vertical ($x$'s match), length $|q - 12 - q| = 12$ (b) horizontal, length $|(m - 4) - (m + 9)| = 13$ (c) vertical, length $c - b$ (d) horizontal, length $r - t$. **6.** (b) $AB = |4 - (-5)| = 9$, $BC = |4 - (-3)| = 7$, $CD = 9$, $DA = 7$. (c) $P = 2(9) + 2(7) = 32$ units. (d) $A = 9 \times 7 = 63$ units^2. **7.** (a) 7, 3, 3, 4, 4, 7 metres, in order around the figure. (b) $7 + 3 + 3 + 4 + 4 + 7 = 28$ m. (c) $28 \times \$6.50 = \$182.00$. **8.** (a) $AB = 9$, $BC = 3$, $CD = 3$, $DE = 2$, $EF = 3$, $FG = 3$, $GH = 9$, $HA = 8$. (b) $9 + 3 + 3 + 2 + 3 + 3 + 9 + 8 = 40$ units. (c) The enclosing rectangle is 9 by 8, so its perimeter is $2(9) + 2(8) = 34$ units — 6 less than the deck. Cutting the notch removes no width or height overall: DE (2) just replaces the piece of the right side that was cut away. The walk around the deck also covers the notch's two inward sides CD and EF (3 each), so the perimeter is 6 more. **9.** $P_1 = 15$, $P_2 = 17$, $P_3 = 14$, $P_4 = 13$, $P_5 = 16$. Shortest to longest: P_4, P_3, P_1, P_5, P_2; P_2 is the longest at 17 units. **10.** (a) $(10, -2)$ or $(-4, -2)$ (b) $(-5, 13)$ or $(-5, -5)$ (c) $(11, -6)$ or $(-13, -6)$ (d) $(7, 17)$ or $(7, -13)$. **11.** (a) $x = 4 - 10 = -6$, so $G(-6, -3)$ (b) $y = 5 + 11 = 16$, so $H(-2, 16)$ (c) $x = -9 - 6 = -15$, so $K(-15, -8)$. **12.** $AB = |9 - (-3)| = 12$, so $2(12) + 2h = 40$, giving $h = 8$. The other two vertices are $(9, 4)$ and $(-3, 4)$ (rectangle above $\overline{AB}$), or $(9, -12)$ and $(-3, -12)$ (rectangle below it). **13.** C — $y = 7 + 15 = 22$ or $y = 7 - 15 = -8$. **14.** (a) She subtracted in the order that gives a negative and then forgot the absolute value. A length measures how far apart two points are, and a distance cannot be negative. Correctly, $RS = |9-(-3)| = 12$ units. (b) The y's matching is what makes $\overline{RS}$ *horizontal*; it is the signal to use the x-coordinates, not something to subtract. He should have computed $|x_2 - x_1| = |9 - (-3)| = 12$ units. **15.** The bars turn the difference into a distance: they strip off the sign and keep the size. Subtracting the other way gives the same number with the opposite sign ($9 - (-3) = 12$ and $-3 - 9 = -12$), so either order is fine once the absolute value is taken — which is exactly why no order is wrong. Length is a measure of how far apart two points are, and there is no such thing as being -12 units apart. **16.** Counterexamples will vary. For instance $(-11, -4)$ to $(-3, -4)$ has length $|-3-(-11)| = 8$ units, which is positive. Negative coordinates only say the points lie left of the y-axis; the length depends on the *gap* between them, not on which side of zero they sit. **17.** Sides $|60-(-40)| = 100$ m and $|35 - (-15)| = 50$ m, so $P = 2(100) + 2(50) = 300$ m; boards cost $300 \times \$85 = \$25\,500$. **18.** One lap $= 2(240) + 2(180) = 840$ m; three laps $= 2520$ m $= 2.52$ km. **19.** (a) $P = 2w + 2h = 30$, so $w + h = 15$. The pairs are $(1, 14)$, $(2, 13)$, $(3, 12)$, $(4, 11)$, $(5, 10)$, $(6, 9)$, $(7, 8)$ and their reverses — 14 ordered pairs, or 7 different rectangle shapes. (b) With $w = 4$, $h = 11$ the rectangle spans 4 units left or right of $x = 2$ and 11 units up or down from $y = 3$. The four rectangles have vertex sets $\{(2, 3), (6, 3), (6, 14), (2, 14)\}$, $\{(2, 3), (-2, 3), (-2, 14), (2, 14)\}$, $\{(2, 3), (6, 3), (6, -8), (2, -8)\}$ and $\{(2, 3), (-2, 3), (-2, -8), (2, -8)\}$. (c) 56. For each of the 14 ordered pairs (w, h) there are 2 choices of horizontal direction and 2 of vertical direction, so $14 \times 2 \times 2 = 56$; no two of these rectangles coincide, because the width and the position of the corner together pin each one down.

Lesson 3 — Extra Practice

Measuring the Length of a Line Segment: Diagonal Segments and the Distance Formula

Part A — Exact Lengths in Simplest Radical Form

Throughout Part A, give each length **twice**: once as an exact value (a mixed radical in simplest form, exactly as in Unit 1, or a whole number if the root is exact), and once as a decimal rounded to the nearest **tenth**. Every pair of points uses negative coordinates.

1. Determine the exact length of each segment, then a decimal approximation.

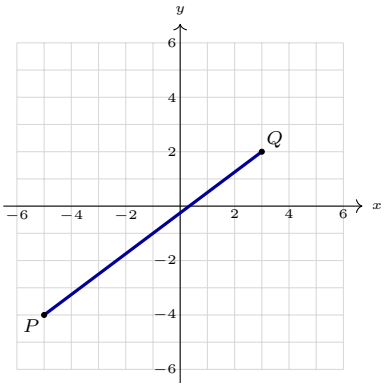
(a) $A(-3,-2)$ to $B(1,4)$ (b) $C(-5,3)$ to $D(2,-1)$ (c) $E(-6,3)$ to $F(0,-5)$

2. Determine the exact length of each segment, then a decimal approximation.

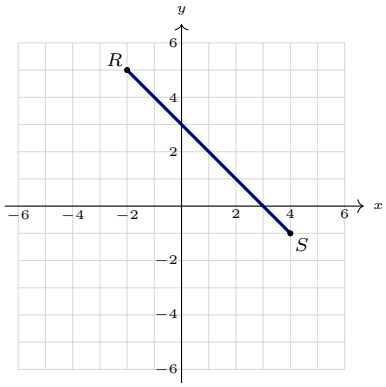
(a) $G(-3,5)$ to $H(1,-3)$ (b) $J(-4,-6)$ to $K(1,-1)$ (c) $L(-4,-3)$ to $M(5,3)$

3. For each segment graphed below: state the coordinates of the two endpoints, sketch in the right triangle underneath the segment, and determine the exact length of the segment.

(a) Segment PQ



(b) Segment RS



4. Four segments are listed below. Two of them have the same length, and the other two also have the same length as each other. Find the exact length of each, then state the two matching pairs.

I: $(-4,1)$ to $(2,5)$ II: $(-3,-5)$ to $(1,1)$ III: $(0,-4)$ to $(5,2)$ IV: $(-6,2)$ to $(-1,-4)$

5. Three points are given: $P(-4, 3)$, $Q(5, -2)$, and $R(1, 6)$. Determine the exact distance from each one to $T(1, -1)$, then state which point is **closest** to T. One of the three distances comes out to a whole number — explain why.

Part B — Classifying Triangles and Quadrilaterals

The distance formula turns a list of vertices into a list of side lengths, and side lengths decide what a figure *is*. Two side lengths equal makes a triangle **isosceles**; all three different makes it **scalene**. To test for a **right angle**, use the **converse of the Pythagorean theorem**: if the two shorter sides squared add to the longest side squared, the triangle is right-angled — and the right angle sits opposite the longest side. Work with the *squared* lengths and you never have to touch a radical.

6. A triangle has vertices $A(-2, -1)$, $B(2, 1)$, and $C(0, 5)$.
(a) Determine the exact length of all three sides. (b) Is the triangle scalene or isosceles?
(c) Use the converse of the Pythagorean theorem on the squared lengths to decide whether the triangle is right-angled. If it is, state which vertex holds the right angle. (d) Name the triangle as completely as you can.

7. A triangle has vertices $P(-5, 1)$, $Q(-1, -2)$, and $R(5, 6)$.
(a) Determine the exact length of all three sides. (b) Is the triangle scalene or isosceles?
(c) Use the converse of the Pythagorean theorem to decide whether the triangle is right-angled, and if so, name the vertex at the right angle. (d) Name the triangle as completely as you can.

8. A triangle has vertices $D(-5, 2)$, $E(-1, -4)$, and $F(3, 2)$.
(a) Determine the exact length of all three sides. (b) Is the triangle scalene or isosceles?
(c) Show, using the squared lengths, that this triangle is **not** right-angled. (d) Name the triangle as completely as you can.

9. A triangle has vertices $A(-4, -3)$, $B(2, -1)$, and $C(-1, 5)$. Every side is diagonal.
(a) Determine the exact length of each side, as a mixed radical in simplest form.
(b) Write the exact perimeter as a sum of radicals.
(c) Determine the perimeter to the nearest tenth.

10. A quadrilateral has vertices $A(-5, -4)$, $B(-1, -1)$, $C(2, 3)$, and $D(-2, 0)$, listed in order.
(a) Determine the length of all four sides and explain why $ABCD$ must be a **rhombus**.
(b) Determine the exact lengths of the two diagonals AC and BD. Since they are not equal, $ABCD$ is *not* a square. Explain.

11. A quadrilateral has vertices $A(-5,-2)$, $B(-1,1)$, $C(4,0)$, and $D(0,-3)$, listed in order. Determine the length of all four sides. Explain why this figure is a **parallelogram** but **not** a rhombus.

Part C — Working Backwards, Errors, and Reasoning

12–13. In each question the distance is known but one coordinate is missing. Substitute into the distance formula, square both sides to clear the radical, and solve. **Each question has two answers** — a squared quantity has two square roots.

12. The distance from $(1,2)$ to $(k,6)$ is 5. Determine both possible values of k, then check each one in the distance formula.

13. The distance from $(-2,3)$ to $(4,k)$ is 10. Determine both possible values of k, then check each one.

14. **Error analysis.** Asked for the distance from $A(-2,-3)$ to $B(4,5)$, Dmitri wrote

$$d = (4-(-2)) + (5-(-3)) = 6+8 = 14.$$

Explain what Dmitri did instead of using the distance formula, and describe the path of length 14 that he actually measured. Then find the correct distance.

15. **Error analysis.** Asked for the distance from $P(-6,2)$ to $Q(-1,-4)$, Lucia wrote

$$d = (-1-(-6))^2 + (-4-2)^2 = 25+36 = 61.$$

Identify the step she left out, state the correct exact distance as a radical, and give it to the nearest tenth.

16. **Reasoning.** Does it matter which point you call (x_1,y_1) and which you call (x_2,y_2)? Compute the distance from $A(-4,5)$ to $B(2,-3)$ *both ways* — once with A first and once with B first — showing the run and rise each time. Then explain, in one or two sentences, why the two calculations must always agree.

17. **Stretch — the perpendicular bisector.** The **perpendicular bisector** of a segment AB is the line of *all* points that are the same distance from A as they are from B. Let $A(-4,-1)$ and $B(2,3)$.

(a) Show that $P(1,-2)$ is the same distance from A as it is from B, by computing both exact distances.

(b) Show that $Q(3,-5)$ is also the same distance from A as it is from B.

(c) The midpoint of AB is $(-1,1)$. Explain why $(-1,1)$, P, and Q must all lie on one straight line, and name that line.

18. **Stretch.** A point on the x-axis has the form $(x, 0)$ — its y-coordinate is 0. Find the point on the x-axis that is the same distance from $A(-4, 2)$ as it is from $B(2, 4)$. *Hint:* write both squared distances in terms of x, set them equal, and solve; the x^2 terms cancel. Then state the common distance as a mixed radical.

Answer Key. 1. (a) $2\sqrt{13} \approx 7.2$ (b) $\sqrt{65} \approx 8.1$ (c) 10. **2.** (a) $4\sqrt{5} \approx 8.9$ (b) $5\sqrt{2} \approx 7.1$ (c) $3\sqrt{13} \approx 10.8$. **3.** (a) $P(-5, -4)$, $Q(3, 2)$; run 8, rise 6, $PQ = 10$ (b) $R(-2, 5)$, $S(4, -1)$; run 6, rise -6, $RS = \sqrt{72} = 6\sqrt{2} \approx 8.5$. **4.** I $= 2\sqrt{13}$, II $= 2\sqrt{13}$, III $= \sqrt{61}$, IV $= \sqrt{61}$; the pairs are I with II and III with IV. **5.** $TP = \sqrt{41} \approx 6.4$, $TQ = \sqrt{17} \approx 4.1$, $TR = 7$; Q is closest. TR is a whole number because T and R share the x-coordinate 1, so the run is 0 and the segment is vertical — the distance is just the difference in y, $6 - (-1) = 7$. **6.** (a) $AB = 2\sqrt{5}$, $BC = 2\sqrt{5}$, $AC = 2\sqrt{10}$ (b) isosceles, since $AB = BC$ (c) squares are 20, 20, 40, and $20 + 20 = 40$, so yes — right-angled, with the right angle at B (opposite the longest side AC) (d) an isosceles right triangle. **7.** (a) $PQ = 5$, $QR = 10$, $PR = 5\sqrt{5}$ (b) scalene — all three differ (c) squares are 25, 100, 125, and $25 + 100 = 125$, so yes, with the right angle at Q (d) a scalene right triangle. **8.** (a) $DE = 2\sqrt{13}$, $EF = 2\sqrt{13}$, $DF = 8$ (b) isosceles, $DE = EF$ (c) the longest side is DF, with $DF^2 = 64$; the two shorter sides give $52 + 52 = 104 \neq 64$, so the converse fails and the triangle is not right-angled. In fact $104 > 64$, so the angle at E is acute (d) an isosceles triangle that is not right-angled. **9.** (a) $AB = 2\sqrt{10}$, $BC = 3\sqrt{5}$, $AC = \sqrt{73}$ (b) $2\sqrt{10} + 3\sqrt{5} + \sqrt{73}$ (c) ≈ 21.6 units. **10.** (a) $AB = BC = CD = DA = 5$; all four sides equal, which is the definition of a rhombus (b) $AC = 7\sqrt{2} \approx 9.9$ and $BD = \sqrt{2} \approx 1.4$. A square is a rhombus whose diagonals are equal; here they are wildly different, so $ABCD$ is a (very "squashed") rhombus, not a square. **11.** $AB = 5$, $BC = \sqrt{26}$, $CD = 5$, $DA = \sqrt{26}$. Both pairs of *opposite* sides are equal ($AB = CD$ and $BC = DA$), so it is a parallelogram; but $5 \neq \sqrt{26}$, so the four sides are not all equal and it is not a rhombus. **12.** $(k-1)^2 + 16 = 25$, so $(k-1)^2 = 9$ and $k - 1 = \pm 3$: $k = 4$ or $k = -2$. **13.** $36 + (k-3)^2 = 100$, so $(k-3)^2 = 64$ and $k - 3 = \pm 8$: $k = 11$ or $k = -5$. **14.** He added the run and the rise instead of squaring them, adding, and taking the square root. His 14 is the length of the "staircase" path — 6 units across, then 8 units up — not the straight-line distance. Correctly, $d = \sqrt{6^2 + 8^2} = \sqrt{100} = 10$. **15.** She never took the square root; 61 is d^2, not d. The distance is $d = \sqrt{61} \approx 7.8$. **16.** A first: run $= 2 - (-4) = 6$, rise $= -3 - 5 = -8$, $d = \sqrt{36 + 64} = 10$. B first: run $= -4 - 2 = -6$, rise $= 5 - (-3) = 8$, $d = \sqrt{36 + 64} = 10$. Reversing the order only flips the sign of each difference, and squaring a number and its opposite gives the same result, so the two calculations can never differ. **17.** (a) $PA = \sqrt{(-4-1)^2 + (-1+2)^2} = \sqrt{26}$ and $PB = \sqrt{(2-1)^2 + (3+2)^2} = \sqrt{26}$; equal. (b) $QA = \sqrt{49 + 16} = \sqrt{65}$ and $QB = \sqrt{1 + 64} = \sqrt{65}$; equal. (c) All three points are equidistant from A and from B, and the set of all such points is exactly the perpendicular bisector of AB — a single straight line. So $(-1, 1)$, P, and Q all lie on the perpendicular bisector of AB. **18.** Setting $(x+4)^2 + 2^2 = (x-2)^2 + 4^2$ gives $x^2 + 8x + 20 = x^2 - 4x + 20$, so $12x = 0$ and $x = 0$. The point is the origin $(0, 0)$, and its distance to each of A and B is $\sqrt{20} = 2\sqrt{5} \approx 4.5$.

Lesson 4 — Extra Practice

The Midpoint Formula

Part A — Midpoints That Land Between the Grid Lines

When the two x-coordinates have *different* parities — one odd, one even — their sum is odd and halving it gives a half-integer such as $-\frac{5}{2}$. That is a perfectly good midpoint: it sits exactly halfway between two grid lines. **Leave these answers as exact fractions (or exact decimals); do not round.**

1. Determine the midpoint of the segment with each pair of endpoints.

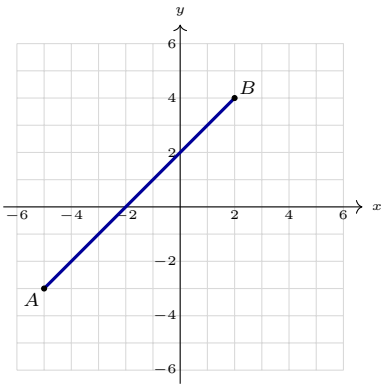
(a) $A(-7, 3),\ B(2, -6)$ (b) $C(-5, -8),\ D(1, 3)$ (c) $E(4, -9),\ F(-3, -1)$

2. Determine the midpoint of the segment with each pair of endpoints. Exactly one of the six coordinates you produce is a whole number — say which one, and explain what has to be true of two numbers for their average to come out whole.

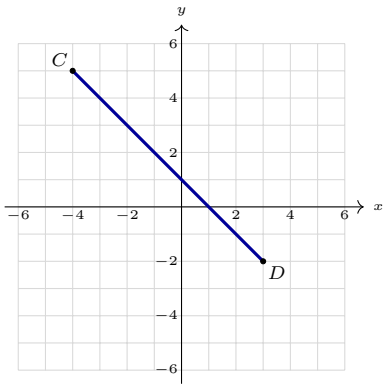
(a) $G(-11, 6),\ H(-2, -3)$ (b) $J(-6, 7),\ K(5, 7)$ (c) $L(-8, -5),\ M(3, -4)$

3. For each segment graphed below: state the coordinates of the two endpoints, determine the midpoint using the formula, and mark it on the grid with a dot. Notice that the dot does *not* land on a lattice point.

(a) Segment AB



(b) Segment CD



4. **Error analysis.** Asked for the midpoint of $A(-6, 5)$ and $B(2, -3)$, Owen wrote

$$M = \left(\frac{2 - (-6)}{2},\ \frac{-3 - 5}{2}\right) = (4, -4).$$

Explain what Owen confused the midpoint formula with, describe what the numbers 4 and -4 actually represent, and give the correct midpoint.

Part B — Running the Formula Backwards

5–6. Each question gives one endpoint A and the midpoint M, and asks for the *other* endpoint B. Two methods work. **Algebra:** solve $\frac{x_A + x_B}{2} = x_M$ for x_B, giving $x_B = 2x_M - x_A$ (and likewise for y). **Counting:** whatever step takes you from A to M, take the same step again to reach B. Use one method and check with the other.

5. Determine the other endpoint B.
(a) $A(-3, 2)$, midpoint $M(1, -1)$
(b) $A(4, -7)$, midpoint $M(-2, -1)$

6. Determine the other endpoint. The midpoints here are half-integers.
(a) $A(-5, -4)$, midpoint $M\left(-\frac{1}{2}, 3\right)$
(b) $P(6, -3)$, midpoint $M\left(\frac{3}{2}, -\frac{11}{2}\right)$

7. The midpoint of AB is $(2, -1)$ and A is $(-3, 4)$.
(a) Determine the coordinates of B. (b) Verify your answer by substituting A and B back into the midpoint formula. (c) Determine the exact length of AB, as a mixed radical in simplest form, and then to the nearest tenth.

8. **The centre of a rectangle.** A rectangle has vertices $A(-4, -3)$, $B(5, -3)$, $C(5, 3)$, and $D(-4, 3)$. A diagonal of a rectangle passes through its centre, so the midpoint of *either* diagonal should be the same point.
(a) Determine the midpoint of diagonal AC.
(b) Determine the midpoint of diagonal BD. Compare with part (a).
(c) Determine the exact length of each diagonal. Are they equal? Should they be?

9. **Stretch.** A quadrilateral has vertices $A(-5, -3)$, $B(-1, 2)$, $C(4, 3)$, and $D(0, -2)$, listed in order, so its diagonals are AC and BD. Show that the two diagonals have the *same* midpoint, which means they bisect each other. Then find all four side lengths and state what kind of quadrilateral $ABCD$ is.

10–11. **Midpoint and distance together.** If M really is the midpoint of AB, then AM must come out to exactly half of AB. That makes a free check on your work: find M, find AM, find AB, and confirm $AB = 2\,AM$.

10. $A(-6, -2)$ and $B(2, 4)$.
(a) Find M. (b) Find AM and MB. (c) Find AB and confirm the halving check.

11. $A(-5, 3)$ and $B(3, -5)$.
(a) Find M. (b) Find the exact value of AM as a mixed radical. (c) Find the exact value of AB and confirm $AB = 2\,AM$.

Part C — Quarter Points, Context, and Reasoning

12. **Stretch — one quarter of the way.** The midpoint formula only ever cuts a segment in *half* — but halving twice gives a quarter. Let $A(-7,-3)$ and $B(5,9)$.

(a) Find M, the midpoint of AB.

(b) Find the midpoint of A and M. This is the point one quarter of the way from A to B.

(c) Find the midpoint of M and B — the point three quarters of the way from A to B.

(d) Find AB and the distance from A to your answer in (b). Is the second exactly one quarter of the first?

13. **Stretch.** $A(-8,-6)$ and $B(8,10)$. Use repeated halving to find the three points that divide AB into **four equal parts**, list them in order from A to B, and verify that the first two are the same distance apart as A is from the first one.

14. **Error analysis.** One endpoint of a segment is $A(-4,1)$ and the midpoint is $M(2,5)$. Asked for the other endpoint, Priya averaged A and M and answered $(-1,3)$. Explain why averaging is the wrong move here, describe what her point $(-1,3)$ actually is, and give the correct other endpoint.

15. A hiking map is drawn on a coordinate grid where each unit is 1 km. The trailhead is at $(-14,8)$ and the lookout is at $(22,-4)$, joined by a straight trail.

(a) A rest stop is to be built exactly halfway along the trail. Give its coordinates.

(b) Determine the exact total length of the trail, as a mixed radical, and to the nearest tenth of a kilometre.

(c) How far is the rest stop from the trailhead? Answer exactly and to the nearest tenth.

16. Two towns sit on a regional map where each unit is 1 km: Ashcroft at $(-9,-5)$ and Bellmere at $(7,3)$. Residents of both towns agree to meet at the point exactly halfway between them.

(a) Determine the coordinates of the meeting point.

(b) Determine the straight-line distance between the two towns, exactly and to the nearest tenth.

(c) How far does each group travel? Explain how you know without recomputing a distance.

17. **Reasoning.** Is the midpoint of a segment *always* strictly between the two endpoints? Can the midpoint ever be equal to one of the endpoints? Justify both answers using the formula $M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$, and describe the one situation in which the midpoint coincides with an endpoint.

18. **Reasoning stretch — which quadrant?**

(a) $A(-6,4)$ lies in quadrant II and $B(2,-8)$ lies in quadrant IV. Find the midpoint of AB and state which quadrant it lies in. Is it in either of the endpoints' quadrants?

(b) Explain how the midpoint of a quadrant II point and a quadrant IV point can land in quadrant III. Then explain why the midpoint of two quadrant III points **must** always be in quadrant III.

Answer Key. 1. (a) $\left(-\frac{5}{2},-\frac{3}{2}\right)$ (b) $\left(-2,-\frac{5}{2}\right)$ (c) $\left(\frac{1}{2},-5\right)$. **2.** (a) $\left(-\frac{13}{2},\frac{3}{2}\right)$ (b) $\left(-\frac{1}{2},7\right)$ (c) $\left(-\frac{5}{2},-\frac{9}{2}\right)$. The one whole number is the y-coordinate 7 in (b), because 7 and 7 are both odd, so their sum 14 is even. In general a midpoint coordinate is whole exactly when the two coordinates being averaged are both even or both odd. **3.** (a) $A(-5,-3)$, $B(2,4)$, $M=\left(-\frac{3}{2},\frac{1}{2}\right)$ (b) $C(-4,5)$, $D(3,-2)$, $M=\left(-\frac{1}{2},\frac{3}{2}\right)$. **4.** He subtracted the coordinates instead of adding them — that is the start of the *distance* formula, not the midpoint formula. His 4 is half the run and his -4 is half the rise: the *step* from one endpoint to the midpoint, not a location. The correct midpoint is $\left(\frac{-6+2}{2},\frac{5+(-3)}{2}\right)=(-2,1)$. **5.** (a) $B(5,-4)$ (b) $B(-8,5)$. **6.** (a) $B(4,10)$ (b) $(-3,-8)$. **7.** (a) $B(7,-6)$ (b) $\left(\frac{-3+7}{2},\frac{4+(-6)}{2}\right)=(2,-1)$ ✓(c) $AB=\sqrt{10^2+(-10)^2}=\sqrt{200}=10\sqrt{2}\approx 14.1$. **8.** (a) $\left(\frac{1}{2},0\right)$ (b) $\left(\frac{1}{2},0\right)$ — identical, so both diagonals pass through the same centre (c) $AC=BD=\sqrt{81+36}=\sqrt{117}=3\sqrt{13}\approx 10.8$; yes they are equal, and they should be — the diagonals of a rectangle are always congruent. **9.** Midpoint of $AC=\left(-\frac{1}{2},0\right)$; midpoint of $BD=\left(-\frac{1}{2},0\right)$ — the same point, so the diagonals bisect each other. Sides: $AB=\sqrt{41}$, $BC=\sqrt{26}$, $CD=\sqrt{41}$, $DA=\sqrt{26}$. Opposite sides are equal and the diagonals bisect each other, so $ABCD$ is a parallelogram (not a rhombus, since $\sqrt{41}\neq\sqrt{26}$). **10.** (a) $M(-2,1)$ (b) $AM=\sqrt{16+9}=5$ and $MB=\sqrt{16+9}=5$ (c) $AB=\sqrt{64+36}=10=2\times 5$ ✓ **11.** (a) $M(-1,-1)$ (b) $AM=\sqrt{16+16}=\sqrt{32}=4\sqrt{2}\approx 5.7$ (c) $AB=\sqrt{64+64}=\sqrt{128}=8\sqrt{2}\approx 11.3$, and $2\times 4\sqrt{2}=8\sqrt{2}$ ✓ **12.** (a) $M(-1,3)$ (b) $(-4,0)$ (c) $(2,6)$ (d) $AB=\sqrt{144+144}=12\sqrt{2}\approx 17.0$ and the distance from $A(-7,-3)$ to $(-4,0)$ is $\sqrt{9+9}=3\sqrt{2}\approx 4.2$; since $4\times 3\sqrt{2}=12\sqrt{2}$, it is exactly one quarter. **13.** $M=(0,2)$; midpoint of A and M is $(-4,-2)$; midpoint of M and B is $(4,6)$. In order from A: $(-4,-2)$, $(0,2)$, $(4,6)$. Check: A to $(-4,-2)$ is $\sqrt{16+16}=4\sqrt{2}$, and $(-4,-2)$ to $(0,2)$ is also $\sqrt{16+16}=4\sqrt{2}$ — equal. (Total $AB=16\sqrt{2}\approx 22.6$, which is $4\times 4\sqrt{2}$.) **14.** Averaging A and M answers a different question — it finds the point halfway between A and M, which is the point one *quarter* of the way from A to B. To go backwards you must *double*: $B=(2x_M-x_A,\,2y_M-y_A)=(2(2)-(-4),\,2(5)-1)=(8,9)$. Check: the midpoint of $(-4,1)$ and $(8,9)$ is $(2,5)$ ✓ **15.** (a) $(4,2)$ (b) $\sqrt{36^2+(-12)^2}=\sqrt{1440}=12\sqrt{10}\approx 37.9$ km (c) $6\sqrt{10}\approx 19.0$ km — exactly half of (b). **16.** (a) $(-1,-1)$ (b) $\sqrt{16^2+8^2}=\sqrt{320}=8\sqrt{5}\approx 17.9$ km (c) each group travels $4\sqrt{5}\approx 8.9$ km; no new calculation is needed, because the meeting point is the midpoint and so is exactly half the total distance from each town. **17.** Yes, provided $A\neq B$. The average of two numbers always lies between them — strictly between when they differ, and equal to both when they are the same (as with the two x-coordinates of a vertical segment). So M always lands on the segment, never past an end. The midpoint equals an endpoint only in the degenerate case where the two "endpoints" are the same point: if $A=B$ then $\frac{x_1+x_2}{2}=x_1$ and $\frac{y_1+y_2}{2}=y_1$, so $M=A=B$ and the segment has length 0. **18.** (a) $M(-2,-2)$, which is in quadrant III — neither endpoint's quadrant. (b) A quadrant II point has $x<0$ and a quadrant IV point has $y<0$; averaging keeps the negative x from the first point if it outweighs the positive x of the second, and keeps a negative y from the second if it outweighs the positive y of the first, so both averages can come out negative — quadrant III. For two quadrant III points, *both* x's are negative and *both* y's are negative; the average of two negatives is always negative, so the midpoint must have $x<0$ and $y<0$ and cannot escape quadrant III.

Lesson 5 — Extra Practice

Applications: Distance, Midpoint, and Finding a Missing Endpoint

Part A — Missing Endpoints and Multi-Step Chains

1. One endpoint and the midpoint of a segment are given. Determine the other endpoint. Watch the signs in (a) and (b); in (c) the midpoint lands on a half-unit.

(a) $A(-7, 4)$ is an endpoint of $\overline{AB}$, with midpoint $M(-2, -3)$. Find B.

(b) $C(5, -11)$ is an endpoint of $\overline{CD}$, with midpoint $M(-1, -4)$. Find D.

(c) $E(3, 8)$ is an endpoint of $\overline{EF}$, with midpoint $M\left(\frac{1}{2}, \frac{5}{2}\right)$. Find F.

2. **Three steps in a row.** $A(1, 2)$ is one endpoint of $\overline{AB}$, and $M(5, 8)$ is the midpoint of $\overline{AB}$.

(a) Determine the coordinates of B.

(b) Determine the exact length of $\overline{AB}$ in simplest radical form, then round to the nearest tenth.

(c) Determine the midpoint of $\overline{MB}$ — the point one-quarter of the way along $\overline{AB}$, measuring from B.

3. **Error analysis.** Devon is told that $A(3, -4)$ is one endpoint of $\overline{AB}$ and that $M(7, 1)$ is the midpoint. He writes

$$B = (7 - 3,\ 1 - (-4)) = (4, 5).$$

(a) Show that Devon's point is not correct by finding the midpoint of $\overline{AB}$ using his answer for B.

(b) Explain what Devon actually calculated, then find the correct coordinates of B.

4. **Work backwards.** A segment has one endpoint at $P(2, 3)$, has a length of 10 units, and its midpoint lies on the x-axis.

(a) Explain why the y-coordinate of the other endpoint must be -3.

(b) Use the distance formula to find **both** possible locations of the other endpoint.

(c) State the midpoint that goes with each of your two answers.

5. **Reasoning.** Consider a segment with endpoints (x_1, y_1) and (x_2, y_2).

(a) What must be true about y_1 and y_2 for the midpoint to land on the x-axis? Give an example.

(b) What must be true about x_1 and x_2 for the midpoint to land on the y-axis? Give an example.

(c) What must be true for the midpoint to be the origin? Give an example.

Part B — Maps, Routes, and Costs

6. On a city map, the library sits at $L(-6, 2)$, the rink at $R(6, 7)$, and the gym at $G(2, -6)$. Each grid unit represents 100 m.

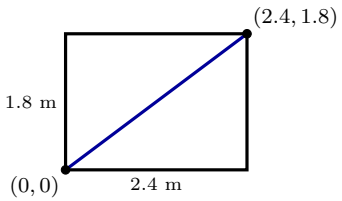
(a) Determine the exact distance between each of the three pairs of buildings. Leave radicals in simplest form.

(b) Which two buildings are closest together? Justify your choice by comparing the three distances to the nearest tenth of a unit.

(c) Two students want to meet exactly halfway between the **closest** pair. Determine the coordinates of that meeting point.

(d) How far, in metres, would each student walk to reach the meeting point? Round to the nearest metre.

7. A rectangular gate is 2.4 m wide and 1.8 m tall. Place the bottom-left corner at $(0, 0)$, so the top-right corner is at $(2.4, 1.8)$. A diagonal brace runs corner to corner.



(a) Use the distance formula to determine the exact length of the brace.

(b) Brace stock costs \$14.50 per metre. Determine the cost of one brace.

(c) The builder decides to add a second brace running from $(0, 1.8)$ to $(2.4, 0)$, forming an X. Determine the total cost of both braces.

8. A courier leaves the depot at $D(-6, -3)$, drives to $A(2, 3)$, then to $B(6, 0)$, then to $C(6, -8)$. Each grid unit represents 500 m.

(a) Determine the length of each of the three legs, in grid units.

(b) Determine the total distance driven, in kilometres.

(c) Determine the straight-line distance from D directly to C, in kilometres.

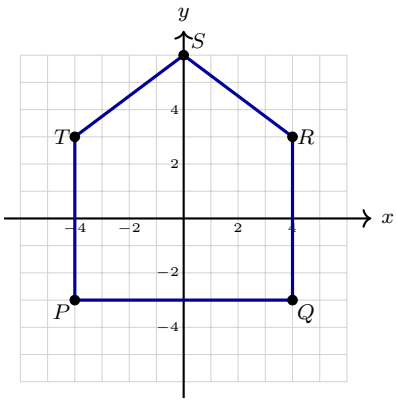
(d) How much shorter is the direct route? Explain why the direct route must be shorter than any route that detours through other stops.

9. Two towns, Ashford at $A(-11, 6)$ and Baxter at $B(5, -6)$, agree to share a cell tower placed exactly halfway between them. Each grid unit represents 2 km.
(a) Determine the coordinates of the tower.
(b) Determine the distance between the two towns, in kilometres.
(c) Determine the distance from Ashford to the tower, in kilometres, and confirm that it is exactly half of your answer to (b).
(d) The tower ends up on one of the axes. Which one, and what does that tell you about the y-coordinates of the two towns?

Part C — Geometry on the Coordinate Grid

10. Quadrilateral $ABCD$ has vertices $A(-6, -1)$, $B(-2, 5)$, $C(6, 3)$, and $D(2, -3)$.
(a) Determine the midpoint of diagonal $\overline{AC}$.
(b) Determine the midpoint of diagonal $\overline{BD}$.
(c) What do your answers to (a) and (b) show about the two diagonals? Use the word **bisect** in your explanation.
(d) Determine the exact lengths of all four sides. What kind of quadrilateral does this confirm $ABCD$ is?

11. A garden plot is the pentagon $PQRST$ shown below, with $P(-4, -3)$, $Q(4, -3)$, $R(4, 3)$, $S(0, 6)$, and $T(-4, 3)$. Each grid unit represents 1 m.



(a) Three of the five sides are parallel to an axis. Name them and state their lengths without using the distance formula.
(b) Use the distance formula on the remaining two sides.
(c) Determine the perimeter of the plot, in metres.
(d) Fencing costs \$23 per metre. Determine the cost of fencing the plot.

12. A triangle has vertices $A(-2,3)$, $B(4,1)$, and $C(0,-3)$.
(a) Determine the exact length of each side in simplest radical form.
(b) Classify the triangle as scalene, isosceles, or equilateral. Justify your answer.
(c) A **median** joins a vertex to the midpoint of the opposite side. Determine the midpoint of $\overline{BC}$, then the exact length of the median drawn from A.

13. The points $(-5,2)$ and $(7,-3)$ are the endpoints of a **diameter** of a circle.
(a) Determine the coordinates of the centre of the circle. (b) Determine the length of the diameter, then the radius.

Part D — Stretch, Multiple Choice, and Numerical Response

14. **Stretch.** A point P lies on the y-axis and is the same distance from $A(-4,1)$ as it is from $B(2,5)$.
(a) Because P is on the y-axis, its coordinates can be written $P(0,k)$. Write an equation saying that the distance PA equals the distance PB. *Hint:* squaring both sides removes both radicals.
(b) Solve your equation for k and state the coordinates of P.
(c) Check your answer by computing both distances.

15. *(Multiple Choice)* One endpoint of a segment is $(-3,8)$ and the midpoint is $(1,2)$. The other endpoint is
A. $(4,-6)$ B. $(5,-4)$ C. $(-1,5)$ D. $(-7,14)$

16. *(Multiple Choice)* The distance between $(-6,-2)$ and $(2,4)$ is
A. $\sqrt{10}$ B. 10 C. 14 D. 100

17. *(Numerical Response)* $M(2,5)$ is the midpoint of $\overline{CD}$ and C is $(-4,-1)$. The x-coordinate of D is _____.

18. *(Numerical Response)* The distance between $(-5,7)$ and $(7,-2)$ is _____ units.

Answer Key. 1. (a) $B(3,-10)$ (b) $D(-7,3)$ (c) $F(-2,-3)$. **2.** (a) $B(9,14)$ (b) $\sqrt{208} = 4\sqrt{13} \approx 14.4$ (c) $(7,11)$. **3.** (a) Using $B(4,5)$ the midpoint would be $(\frac{3+4}{2}, \frac{-4+5}{2}) = (3.5, 0.5)$, not $M(7,1)$. (b) He found the *change* from A to M (right 4, up 5) instead of the endpoint; that change must be applied *again* starting at M, so $B = (7+4, 1+5) = (11,6)$ — equivalently $B = (2\cdot 7 - 3, 2\cdot 1 - (-4)) = (11,6)$. **4.** (a) The midpoint's y-coordinate is $\frac{3+y_2}{2}$; setting it to 0 gives $y_2 = -3$. (b) $\sqrt{(x_2-2)^2+36} = 10 \Rightarrow (x_2-2)^2 = 64 \Rightarrow x_2 = 10$ or $x_2 = -6$, so $(10,-3)$ or $(-6,-3)$. (c) Midpoints $(6,0)$ and $(-2,0)$ respectively. **5.** (a) y_1 and y_2 must be opposites, i.e. $y_1 + y_2 = 0$; e.g. $(1,4)$ and $(7,-4)$ give $M(4,0)$. (b) $x_1 + x_2 = 0$; e.g. $(-3,1)$ and $(3,9)$ give $M(0,5)$. (c) Both conditions at once: $x_1 + x_2 = 0$ *and* $y_1 + y_2 = 0$; e.g. $(5,-2)$ and $(-5,2)$. **6.** (a) $LR = 13$, $LG = \sqrt{128} = 8\sqrt{2}$, $RG = \sqrt{185}$. (b) $LG \approx 11.3$, $LR = 13.0$, $RG \approx 13.6$, so the library and the gym are closest. (c) $M(-2,-2)$. (d) Half of $8\sqrt{2}$ is $4\sqrt{2} \approx 5.657$ units, so about 566 m each. **7.** (a) $\sqrt{2.4^2 + 1.8^2} = \sqrt{9} = 3$ m. (b) $3 \times \$14.50 = \$43.50$. (c) The second brace has the same length, so \$87.00. **8.** (a) $DA = 10$, $AB = 5$, $BC = 8$. (b) 23 units = 11.5 km. (c) $DC = \sqrt{144+25} = 13$ units = 6.5 km. (d) 5 km shorter. A straight line is the shortest path between two points, so any detour adds distance. **9.** (a) $(-3,0)$. (b) $\sqrt{256+144} = 20$ units = 40 km. (c) $\sqrt{64+36} = 10$ units = 20 km, exactly half. (d) The x-axis; the towns' y-coordinates (6 and -6) are opposites, so they add to zero. **10.** (a) $(0,1)$. (b) $(0,1)$. (c) The diagonals share the same midpoint, so each one cuts the other exactly in half — the diagonals bisect each other. (d) $AB = CD = \sqrt{52} = 2\sqrt{13}$ and $BC = DA = \sqrt{68} = 2\sqrt{17}$; opposite sides are equal, so $ABCD$ is a parallelogram. **11.** (a) $\overline{PQ}$ is horizontal with length 8 m; $\overline{QR}$ and $\overline{TP}$ are vertical, each of length 6 m. (b) $RS = \sqrt{16+9} = 5$ m and $ST = \sqrt{16+9} = 5$ m. (c) $8+6+5+5+6 = 30$ m. (d) $30 \times \$23 = \$690$. **12.** (a) $AB = \sqrt{40} = 2\sqrt{10}$, $BC = \sqrt{32} = 4\sqrt{2}$, $CA = \sqrt{40} = 2\sqrt{10}$. (b) Isosceles: $AB = CA$ but BC differs. (c) Midpoint of

$\overline{BC}$ is $(2,-1)$; the median from A has length $\sqrt{16+16}=\sqrt{32}=4\sqrt{2}$. **13.** (a) $\left(1,-\frac{1}{2}\right)$. (b) Diameter $=\sqrt{144+25}=13$; radius $=\frac{13}{2}=6.5$. **14.** (a) $\sqrt{(0+4)^2+(k-1)^2}=\sqrt{(0-2)^2+(k-5)^2}$, so $16+(k-1)^2=4+(k-5)^2$. (b) $17-2k=29-10k \Rightarrow 8k=12 \Rightarrow k=\frac{3}{2}$, so $P\left(0,\frac{3}{2}\right)$. (c) $PA=\sqrt{16+\frac{1}{4}}=\sqrt{\frac{65}{4}}=\frac{\sqrt{65}}{2}$ and $PB=\sqrt{4+\frac{49}{4}}=\sqrt{\frac{65}{4}}=\frac{\sqrt{65}}{2}$. ✓ **15.** B. **16.** B. **17.** 8. **18.** 15.

Lesson 6 — Extra Practice

Rise Over Run: The Slope of a Line Segment

A head start on Lesson 7: some of these segments fall to the right (negative slope) or are flat or vertical. Use the same formula and watch the signs.

Part A — The Slope Formula with Negative Coordinates

1. Determine the slope of the segment joining each pair of points. Every answer here is a **fraction in lowest terms** or a whole number — never a decimal.

(a) $A(-7,5),\ B(2,-1)$ (b) $C(-3,-8),\ D(5,-2)$ (c) $E(4,-6),\ F(-8,3)$

(d) $G(-11,-2),\ H(1,-8)$ (e) $J(-6,-9),\ K(4,-4)$ (f) $L(5,-8),\ N(-4,4)$

2. Determine each slope. Two of these six are special cases — name them rather than writing a fraction.

(a) $P(-5,7),\ Q(3,7)$ (b) $R(-2,-4),\ S(-2,6)$ (c) $T(-8,5),\ U(2,-10)$

(d) $V(-9,4),\ W(6,-1)$ (e) $X(6,-11),\ Y(-4,-3)$ (f) $Z(-10,7),\ B(4,-7)$

3. **Zero versus undefined.** Segment $\overline{JK}$ joins $J(-6,-3)$ and $K(7,-3)$. Segment $\overline{LN}$ joins $L(4,-7)$ and $N(4,5)$.

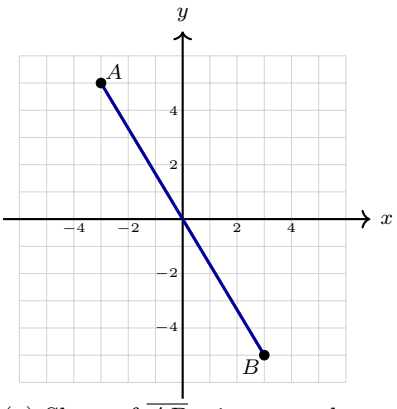
(a) Compute the rise and the run for $\overline{JK}$, then its slope.

(b) Compute the rise and the run for $\overline{LN}$, then its slope.

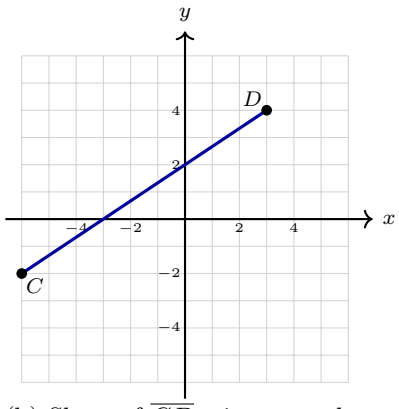
(c) One of these slopes is 0 and the other is **undefined**. Explain the difference by referring to which of rise and run is zero, and why dividing by zero is not allowed.

Part B — Reading Slope from a Graph

4. For each segment below, count the rise and the run from the labelled point on the left to the labelled point on the right, then state the slope in lowest terms. One of the two slopes is negative.

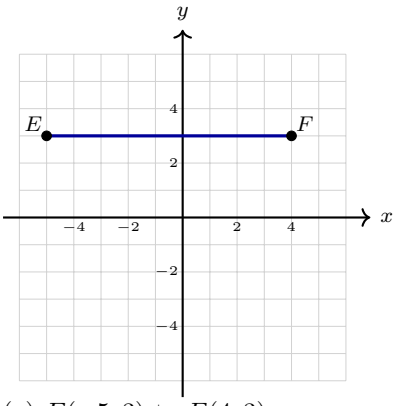


(a) Slope of $\overline{AB}$: rise, run, slope.

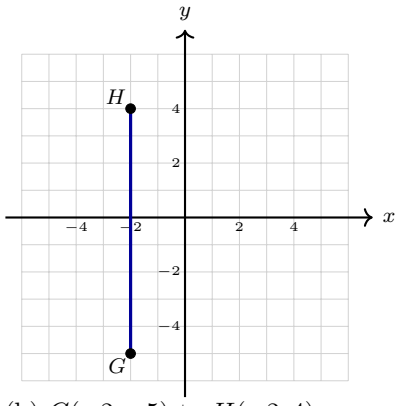


(b) Slope of $\overline{CD}$: rise, run, slope.

5. The two segments below are the same two special types you met in question 3. State the rise, the run, and the slope of each, and say which one has no slope value at all.



(a) $E(-5, 3)$ to $F(4, 3)$.



(b) $G(-2, -5)$ to $H(-2, 4)$.

6. **Which is steepest?** Five segments are described by their endpoints.
$\overline{AB}$: $A(-4, -1)$, $B(4, 5)$ $\overline{CD}$: $C(-2, 6)$, $D(2, -4)$ $\overline{EF}$: $E(-5, -6)$, $F(-1, 2)$
$\overline{GH}$: $G(-6, 3)$, $H(6, -1)$ $\overline{JK}$: $J(0, -5)$, $K(4, 2)$
(a) Determine the five slopes in lowest terms.
(b) Order the segments from **steepest to least steep**. *Careful:* steepness depends on how far a slope sits from zero, not on whether it is positive or negative.
(c) Which segments fall from left to right? How do you know from the slope alone?

Part C — Slope as a Rate of Change

7. A ramp's **grade** is its slope, usually reported as a percent.
(a) A ramp rises 0.45 m over a horizontal run of 5.4 m. Determine its grade as a fraction in lowest terms, then as a percent rounded to the nearest tenth.

(b) A driveway rises 3 m over a horizontal run of 25 m. Determine its grade as a fraction in lowest terms and as an exact percent. Which of (a) and (b) is steeper?

8. A print shop charges by the poster. An order of 4 posters costs \$47, and an order of 10 posters costs \$110. Treat the points as $(\text{posters}, \text{cost})$.
(a) Determine the slope of the segment joining $(4, 47)$ and $(10, 110)$, as a fraction in lowest terms.
(b) State the **units** of that slope, and write one sentence explaining what the number means for this print shop.
(c) Using the same rate, how much more would an order of 16 posters cost than an order of 10?

9. A cyclist's distance from a marker is recorded at two times: at 3 s she is 14 m away, and at 11 s she is 50 m away. Treat the points as $(\text{time in s}, \text{distance in m})$.
(a) Determine the slope of the segment joining $(3, 14)$ and $(11, 50)$ as a fraction in lowest terms.

(b) State the units of the slope and explain what it tells you about the cyclist. Why is a slope with units of metres per second called a *speed*?

10. A tank is draining. After 5 minutes it holds 340 L; after 20 minutes it holds 115 L.
(a) Determine the slope of the segment joining $(5, 340)$ and $(20, 115)$.
(b) State the units of the slope and explain what the **negative sign** means in this context.
(c) If the tank keeps draining at this rate, how much water is left after 25 minutes?

Part D — Working Backwards, Errors, and Reasoning

11. In each part, the slope is known but something else is missing. Set up the slope formula and solve.

(a) The slope from $(2, 5)$ to $(6, y)$ is $\frac{3}{4}$. Find y.

(b) The slope from $(-3, k)$ to $(5, 7)$ is $-\frac{1}{2}$. Find k.

(c) The slope from $(-4, 3)$ to $(x, -1)$ is $\frac{2}{3}$. Find x.

(d) The slope from $(1, -6)$ to $(x, 4)$ is $-\frac{5}{2}$. Find x.

(e) A line passes through $(1, 2)$ with slope -2. Find two other points on it.

(f) A line passes through $(-4, -1)$ and has an **undefined** slope. Find two other points on it.

12. **Error analysis.** Priya computes the slope of the segment joining $(-2, 1)$ and $(6, 7)$ and writes $m = \dfrac{6-(-2)}{7-1} = \dfrac{8}{6} = \dfrac{4}{3}$.

(a) Name the mistake precisely and state the correct slope.

(b) A slope of $\frac{4}{3}$ is steeper than the correct answer. Explain why flipping rise and run always exaggerates a slope that should be less than 1.

13. **Error analysis.** Marcus computes the slope of the segment joining $(-5, 2)$ and $(3, -4)$ and writes $m = \dfrac{-4-2}{-5-3} = \dfrac{-6}{-8} = \dfrac{3}{4}$.

(a) His arithmetic is correct, but his answer is wrong. Explain exactly which rule he broke.

(b) State the correct slope, and say what the sign of the correct answer tells you about the direction of the segment.

14. **Reasoning.** Does it matter which point you call "first"? Use $(-5, 2)$ and $(3, -4)$.

(a) Compute the slope treating $(-5, 2)$ as (x_1, y_1).

(b) Compute the slope treating $(3, -4)$ as (x_1, y_1).

(c) Compare the two results and explain, using the idea of a negative divided by a negative, why the choice can never change the answer — provided you subtract in the **same order** on top and bottom.

15. **Stretch — collinear points.** Three points are **collinear** when they all lie on one straight line, which happens exactly when the slope between the first two equals the slope between the last two.

(a) $A(-6,-7)$, $B(2,-1)$, and $C(10,k)$ are collinear. Find the slope of $\overline{AB}$, then use it to find k.

(b) $P(-3,4)$, $Q(1,y)$, and $R(9,-8)$ are collinear. Find the slope of $\overline{PR}$, then use it to find y.

16. *(Multiple Choice)* The slope of the segment joining $(-6,5)$ and $(2,-1)$ is

A. $-\frac{4}{3}$ B. $-\frac{3}{4}$ C. $\frac{3}{4}$ D. $\frac{4}{3}$

17. *(Multiple Choice)* Which point, together with $(-3,-2)$, forms a segment with a slope of $-\frac{3}{5}$?

A. $(2,-5)$ B. $(2,1)$ C. $(-8,3)$ D. $(-3,3)$

18. *(Numerical Response)* The slope of the segment joining $(-9,-4)$ and $(6,-14)$ is $-\frac{a}{b}$ in lowest terms. The value of $a+b$ is _____.

19. *(Numerical Response)* The slope from $(2,-3)$ to $(10,y)$ is $\frac{3}{4}$. The value of y is _____.

Answer Key. 1. (a) $-\frac{2}{3}$ (b) $\frac{3}{4}$ (c) $-\frac{3}{4}$ (d) $-\frac{1}{2}$ (e) $\frac{1}{2}$ (f) $-\frac{4}{3}$. **2.** (a) 0 (horizontal) (b) undefined (vertical) (c) $-\frac{3}{2}$ (d) $-\frac{1}{3}$ (e) $-\frac{4}{5}$ (f) -1. **3.** (a) Rise 0, run 13, slope $\frac{0}{13}=0$. (b) Rise 12, run 0, slope $\frac{12}{0}$, which is undefined. (c) A slope of 0 means the rise is zero — the segment is level, so it climbs nothing over a real horizontal distance. An undefined slope means the *run* is zero — there is no horizontal distance at all, and division by zero has no value, so a vertical segment simply has no slope number. **4.** (a) Rise -10, run 6, slope $\frac{-10}{6}=-\frac{5}{3}$. (b) Rise 6, run 9, slope $\frac{6}{9}=\frac{2}{3}$. **5.** (a) Rise 0, run 9, slope 0. (b) Rise 9, run 0, slope undefined — the vertical segment $\overline{GH}$ is the one with no slope value. **6.** (a) $AB=\frac{6}{8}=\frac{3}{4}$, $CD=\frac{-10}{4}=-\frac{5}{2}$, $EF=\frac{8}{4}=2$, $GH=\frac{-4}{12}=-\frac{1}{3}$, $JK=\frac{7}{4}$. (b) By distance from zero: $|-\frac{5}{2}|=2.5$, $|2|=2$, $|\frac{7}{4}|=1.75$, $|\frac{3}{4}|=0.75$, $|-\frac{1}{3}|\approx 0.33$, so $\overline{CD}$, $\overline{EF}$, $\overline{JK}$, $\overline{AB}$, $\overline{GH}$. (c) $\overline{CD}$ and $\overline{GH}$; their slopes are negative, meaning y decreases as x increases. **7.** (a) $\frac{0.45}{5.4}=\frac{45}{540}=\frac{1}{12}\approx 8.3\%$. (b) $\frac{3}{25}=12\%$; the driveway in (b) is steeper, since $12\%>8.3\%$. **8.** (a) $\frac{110-47}{10-4}=\frac{63}{6}=\frac{21}{2}$. (b) Dollars per poster: each additional poster adds \$10.50 to the bill. (c) 6 extra posters at \$10.50 each = \$63 more. **9.** (a) $\frac{50-14}{11-3}=\frac{36}{8}=\frac{9}{2}$. (b) Metres per second: she moves 4.5 m farther from the marker every second. A slope in metres per second compares distance travelled to time taken, and that ratio is exactly what speed means. **10.** (a) $\frac{115-340}{20-5}=\frac{-225}{15}=-15$. (b) Litres per minute: the tank *loses* 15 L each minute, and the negative sign records that the volume is decreasing. (c) 5 more minutes past 20 removes 75 L, leaving $115-75=40$ L. **11.** (a) $\frac{y-5}{4}=\frac{3}{4}\Rightarrow y-5=3\Rightarrow y=8$. (b) $\frac{7-k}{8}=-\frac{1}{2}\Rightarrow 7-k=-4\Rightarrow k=11$. (c) $\frac{-4}{x+4}=\frac{2}{3}\Rightarrow 2(x+4)=-12\Rightarrow x=-10$. (d) $\frac{10}{x-1}=-\frac{5}{2}\Rightarrow -5(x-1)=20\Rightarrow x=-3$. (e) From $(1,2)$, down 2 for every 1 right: $(2,0)$ and $(3,-2)$; going left instead gives $(0,4)$. (Any two correct points.) (f) An undefined slope means a vertical line, so every point has $x=-4$: e.g. $(-4,0)$ and $(-4,7)$. **12.** (a) She put the run on top and the rise on the bottom — she computed $\frac{x_2-x_1}{y_2-y_1}$ instead of $\frac{y_2-y_1}{x_2-x_1}$. The correct slope is $\frac{7-1}{6-(-2)}=\frac{6}{8}=\frac{3}{4}$. (b) A slope less than 1 has a numerator smaller than its denominator; flipping it puts the larger number on top, producing a value greater than 1 — steeper than the segment really is. **13.** (a) He subtracted in *opposite* orders: the numerator used $(-4)-2$ (second point first) while the denominator used $(-5)-3$ (first point first). Both subtractions must start from the same point. (b) Correct slope $=\frac{-4-2}{3-(-5)}=\frac{-6}{8}=-\frac{3}{4}$. The negative sign means the segment falls as you move right. **14.** (a) $\frac{-4-2}{3-(-5)}=\frac{-6}{8}=-\frac{3}{4}$. (b) $\frac{2-(-4)}{-5-3}=\frac{6}{-8}=-\frac{3}{4}$. (c) The two agree. Swapping the points negates both the rise and the run, and a negative divided by a negative gives the same quotient, so the slope is unchanged — but only if both subtractions are reversed together. **15.** (a) Slope of $\overline{AB}=\frac{-1-(-7)}{2-(-6)}=\frac{6}{8}=\frac{3}{4}$; then $\frac{k-(-1)}{10-2}=\frac{3}{4}\Rightarrow k+1=6\Rightarrow k=5$. (b) Slope of $\overline{PR}=\frac{-8-4}{9-(-3)}=\frac{-12}{12}=-1$; then $\frac{y-4}{1-(-3)}=-1\Rightarrow y-4=-4\Rightarrow y=0$. **16.** B. **17.** A. **18.** 5 (the slope is $-\frac{2}{3}$, so $a+b=2+3=5$). **19.** 3.

Lesson 7 — Extra Practice

Types of Slope: Positive, Negative, Zero, and Undefined

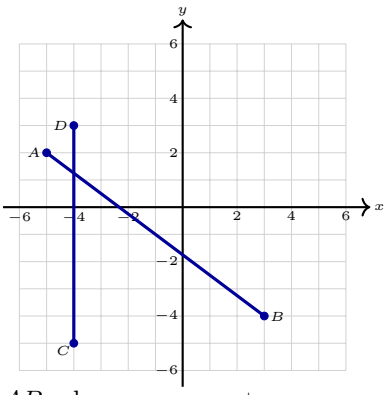
Part A — Naming the Type from All Four Representations

A line's slope type can be read from coordinates, from a picture, from an equation, or from a sentence. The four questions below give you the same job four different ways.

1. **From two points.** State the rise, the run, the slope, and the type (positive, negative, zero, or undefined).
(a) $(-6, 5)$ and $(2, 5)$ (b) $(-3, -7)$ and $(-3, 4)$ (c) $(-5, 8)$ and $(3, -4)$ (d) $(-7, -6)$ and $(5, -1)$

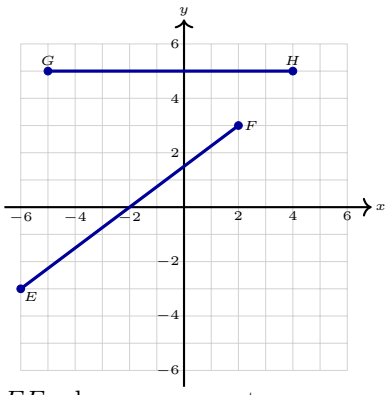
2. **From a graph.** Each grid shows two segments. For each segment, read the endpoints off the grid, calculate the slope, and state its type.

(a)



AB: slope _____ type _____
CD: slope _____ type _____

(b)



EF: slope _____ type _____
GH: slope _____ type _____

3. **From an equation** *(a preview of Unit 7).* In $y = mx + b$, the number multiplying x is the slope; $y = k$ and $x = k$ are the horizontal and vertical lines. Rearrange each equation if you need to, then state the slope and its type.
(a) $y = -4x + 9$ (b) $2x - 10 = 0$ (c) $y - 6 = 0$ (d) $5y = 3x - 15$

4. **From a description.** State the slope type described. Where enough information is given, also state the slope itself.
(a) As x increases by 3, y decreases by 7. (b) Every point on the line has a y-coordinate of -2. (c) The line is parallel to the y-axis. (d) As x increases by 10, y increases by 1.

Part B — Steepness Is Not the Same Thing as Sign

The **sign** of a slope tells you which way the line tilts. The **steepness** tells you how hard the climb is, and steepness ignores the sign entirely — it is the size of the number, the distance from 0. A slope of -4 tilts downward, but it is a very steep downward tilt: it is four times as steep as a slope of $+1$.

5. Four lines have slopes -4, $\frac{1}{2}$, $-\frac{1}{3}$, and 3.

(a) List the four slopes in order of **steepness**, from least steep to steepest.

(b) List the same four slopes in order of **value**, from least to greatest.

(c) Your two lists are not the same. Explain exactly what makes them different.

6. Five lines have slopes $-\frac{5}{2}$, $\frac{7}{4}$, -1, $\frac{1}{5}$, and 2.

(a) List them from **steepest to least steep**.

(b) List them from **greatest value to least value**.

7. For each pair, state which slope belongs to the **steeper** line. Careful — in every pair one slope is negative.

(a) $\frac{3}{4}$ or $-\frac{4}{5}$ (b) -2 or $\frac{9}{4}$ (c) $\frac{5}{6}$ or $-\frac{6}{7}$ (d) $-\frac{7}{2}$ or 3

8. **Error analysis.** Dev is comparing a line of slope -5 with a line of slope 2. He writes: "Since $-5 < 2$, the line with slope -5 is less steep."

(a) Is his inequality $-5 < 2$ correct? Is his conclusion correct? Explain what he has confused.

(b) Sketch a rough picture of both lines through the same point, and use it to say which really is steeper and by how many times.

Part C — Zero and Undefined: The Pair Students Mix Up

9. Complete the table, then answer the question underneath it.

Line	Rise	Run	Slope
horizontal			
vertical			

Also give a sample equation for each: horizontal _____, vertical _____.

Using the words *rise*, *run*, and *divide*, explain why one of these two slopes is the number 0 while the other is not a number at all.

10. **Work backwards.** Give a pair of points that satisfies each requirement, then show the slope calculation that proves it.

(a) Two points whose segment has an **undefined** slope, one of them $(-3, 5)$.

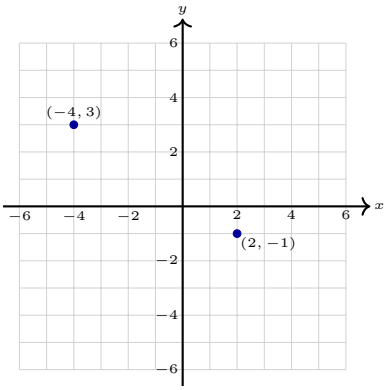
(b) Two points whose segment has slope **zero**, one of them $(-3, 5)$.

(c) Two points whose segment has **negative** slope, with both points in Quadrant II.

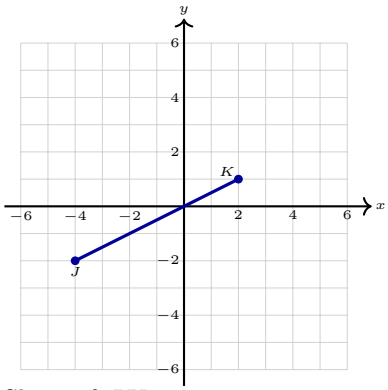
(d) Two points whose segment has **positive** slope, with both y-coordinates negative.

11. **Sketching.** Use the grids below. Draw each segment long enough to read, and label it.

(a) Draw segment PQ with slope 0 passing through $(-4, 3)$, and segment RS with an undefined slope passing through $(2, -1)$.



(b) Segment JK is already drawn. Find its slope, then draw segment LM with a *positive* slope that is *steeper* than JK, and segment NP with a *negative* slope that is *less steep* than JK.



Slope of JK: _____

12. **Error analysis.** Asked for the slope of the segment joining $(4, -1)$ and $(4, 6)$, Priya wrote $m = \dfrac{0}{7} = 0$ and called the line horizontal.

(a) Which number did she put on top, and which should have gone there? Give the correct rise and run.

(b) State the correct slope and the correct description of the line, and give its equation.

Part D — What Slope Means in the Real World

13. A highway sign warns of an 8% downhill **grade**. Grade is slope written as a percent: an 8% downhill grade means the road drops 8 m of elevation for every 100 m of horizontal distance.

(a) Write this grade as a slope, as a fraction in lowest terms and as a decimal. Why is it negative?

(b) How much elevation does a truck lose over a horizontal distance of 2 km?

(c) A second sign reads 11%. Which hill is steeper, and what would a driver physically notice?

14. A wheelchair ramp must not be steeper than a 1:12 slope — that is, at most 1 unit of rise for every 12 units of run. A contractor builds a ramp that rises 0.75 m over a horizontal run of 8 m.

(a) Calculate the ramp's slope as a fraction in lowest terms.

(b) Compare it to $\frac{1}{12}$. Does the ramp meet the requirement? Show how you compared them.

(c) What is the shortest run that would make a 0.75 m rise legal?

15. A roof's **pitch** is written as rise:run. A "5:12 roof" rises 5 ft for every 12 ft of horizontal run.

(a) Write the slopes of a 5:12 roof and a 7:12 roof. Which roof is steeper?

(b) A gable roof has two sides meeting at a peak. If the left side has slope $\frac{7}{12}$, what is the slope of the right side?

(c) The two sides have different slopes but the same steepness. Explain how both statements can be true at once.

16. A tank is draining. Its water depth d (in centimetres) after t minutes is a linear relation with slope -3. At $t = 0$ the depth is 45 cm.

(a) What does the *negative sign* tell you about the water?

(b) What do the *units* of the number 3 tell you? Include the units.

(c) What is the depth after 8 minutes?

(d) When is the tank empty? What happens to the graph after that moment?

Part E — Reasoning and Stretch

17. State whether each claim is **true** or **false**. If it is false, give a specific counterexample — an actual line that breaks the claim.

(a) Every line with a positive slope passes through Quadrant I.

(b) A line with slope 0 passes through exactly two quadrants.

(c) A line with an undefined slope can never pass through Quadrant III.

(d) A line of slope 4 and a line of slope -4 are equally steep.

18. **Stretch.** A segment has one endpoint pinned at the origin. Its other endpoint starts at $(6, 0)$ and rotates counterclockwise about the origin, staying 6 units away, until it finally arrives at $(0, 6)$.

(a) What is the slope of the segment at the very start? _____ What type is it? _____

(b) Partway through the turn, the segment's rise equals its run. What is its slope there, and what fraction of the quarter-turn has it completed?

(c) As the endpoint keeps climbing toward $(0, 6)$, does the slope keep growing, level off, or start shrinking? Does it ever reach a largest possible value?

(d) What is the slope at the very end of the turn? A classmate says "the slope climbs all the way to infinity and then becomes undefined." Explain why calling infinity a slope is not correct, even though his picture of the motion is right.

Answer Key. 1. (a) rise 0, run 8, $m = 0$, zero (b) rise 11, run 0, m undefined (c) rise -12, run 8, $m = -\frac{3}{2}$, negative (d) rise 5, run 12, $m = \frac{5}{12}$, positive. **2.** (a) AB: $A(-5, 2)$, $B(3, -4)$, $m = \frac{-6}{8} = -\frac{3}{4}$, negative; CD: $C(-4, -5)$, $D(-4, 3)$, run $= 0$, m undefined. (b) EF: $E(-6, -3)$, $F(2, 3)$, $m = \frac{6}{8} = \frac{3}{4}$, positive; GH: $G(-5, 5)$, $H(4, 5)$, $m = \frac{0}{9} = 0$, zero. **3.** (a) $m = -4$, negative (b) $x = 5$, vertical, undefined (c) $y = 6$, horizontal, $m = 0$ (d) $y = \frac{3}{5}x - 3$, $m = \frac{3}{5}$, positive. **4.** (a) negative, $m = -\frac{7}{3}$ (b) zero, $m = 0$ (horizontal, $y = -2$) (c) undefined (vertical) (d) positive, $m = \frac{1}{10}$. **5.** (a) steepness $\frac{1}{3}, \frac{1}{2}, 3, 4$, so $-\frac{1}{3}$, $\frac{1}{2}$, 3, -4. (b) -4, $-\frac{1}{3}$, $\frac{1}{2}$, 3. (c) Steepness ignores the sign — it measures only how far the number is from 0 — while value does not. So -4 is the *smallest* value but the *steepest* line, and it moves from one end of the list to the other. **6.** (a) steepness $\frac{5}{2}, 2, \frac{7}{4}, 1, \frac{1}{5}$, so $-\frac{5}{2}$, 2, $\frac{7}{4}$, -1, $\frac{1}{5}$. (b) 2, $\frac{7}{4}$, $\frac{1}{5}$, -1, $-\frac{5}{2}$. **7.** (a) $-\frac{4}{5}$ ($\frac{4}{5} > \frac{3}{4}$) (b) $\frac{9}{4}$ ($\frac{9}{4} > 2$) (c) $-\frac{6}{7}$ ($\frac{6}{7} > \frac{5}{6}$) (d) $-\frac{7}{2}$ ($\frac{7}{2} > 3$). **8.** (a) The inequality is correct but the conclusion is not: he compared *values* when the question asked about *steepness*, which uses the size of the number, $|-5| = 5$ and $|2| = 2$. (b) The slope -5 line is steeper — it is $\frac{5}{2}$ times as steep — but it falls left to right while the slope 2 line rises. **9.** Horizontal: rise 0, run any non-zero number, slope 0, e.g. $y = 4$. Vertical: rise any non-zero number, run 0, slope undefined, e.g. $x = 4$. *Explanation:* slope *divides* the *rise* by the *run*. A horizontal line has rise 0, and 0 divided by anything non-zero is the perfectly ordinary number 0. A vertical line has run 0, and dividing by 0 produces no number at all — so "undefined" is not a strange value, it is the absence of one. **10.** Answers vary. (a) any $(-3, y)$ with $y \neq 5$, e.g. $(-3, 5)$ and $(-3, -2)$: $m = \frac{-7}{0}$, undefined. (b) any $(x, 5)$ with $x \neq -3$, e.g. $(-3, 5)$ and $(4, 5)$: $m = \frac{0}{7} = 0$. (c) both points need $x < 0$, $y > 0$, e.g. $(-5, 6)$ and $(-1, 2)$: $m = -1$. (d) both y-values negative with the right-hand point higher, e.g. $(-2, -7)$ and $(3, -2)$: $m = 1$. **11.** (a) PQ is any horizontal segment along $y = 3$ through $(-4, 3)$; RS is any vertical segment along $x = 2$ through $(2, -1)$. (b) $m_{JK} = \frac{1-(-2)}{2-(-4)} = \frac{3}{6} = \frac{1}{2}$. LM needs any slope greater than $\frac{1}{2}$ (e.g. 2); NP needs a negative slope whose size is less than $\frac{1}{2}$ (e.g. $-\frac{1}{3}$) — note -2 would be wrong, since it is steeper. **12.** (a) She put the change in x on top. The rise is $6 - (-1) = 7$ and the run is $4 - 4 = 0$, so the 0 belongs underneath. (b) $m = \frac{7}{0}$, which is undefined; the line is *vertical*, with equation $x = 4$. **13.** (a) $-\frac{8}{100} = -\frac{2}{25} = -0.08$; negative because elevation decreases as horizontal distance increases. (b) $-0.08 \times 2000 = -160$ m, a drop of 160 m. (c) The 11% hill is steeper; the driver would feel a faster drop and need lower gears or more braking over the same distance. **14.** (a) $\frac{0.75}{8} = \frac{75}{800} = \frac{3}{32}$. (b) $\frac{3}{32} = 0.09375$ and $\frac{1}{12} \approx 0.0833$ (or compare $\frac{9}{96}$ with $\frac{8}{96}$), so the ramp is steeper than allowed — it does **not** meet the requirement. (c) Run $= 12 \times 0.75 = 9$ m. **15.** (a) $\frac{5}{12}$ and $\frac{7}{12}$; the 7:12 roof is steeper. (b) $-\frac{7}{12}$. (c) The two sides tilt in opposite directions, so their slopes have opposite signs and are different numbers; steepness ignores sign, and $\left|\frac{7}{12}\right| = \left|-\frac{7}{12}\right|$, so the climb is equally hard on both sides. **16.** (a) The depth is decreasing — the tank is losing water. (b) The water level drops 3 cm every minute. (c) $45 - 3(8) = 21$ cm. (d) $45 \div 3 = 15$ minutes; after that the relation stops — the depth stays at 0 rather than going negative, so the straight line no longer models the tank. **17.** (a) True. With $m > 0$ the line rises forever to the right, so far enough right both x and y are positive. (b) False. The x-axis, $y = 0$, has slope 0 and passes through no quadrant at all (every other horizontal line does pass through exactly two). (c) False. $x = -3$ has an undefined slope and runs straight down through Quadrant II into Quadrant III. (d) True. Steepness is the size of the slope, and $|4| = |-4| = 4$; they tilt opposite ways but climb equally hard. **18.** (a) 0; zero slope (the segment lies flat along the positive x-axis). (b) Slope $\frac{\text{rise}}{\text{run}} = 1$; that is halfway through the quarter-turn (45° of the 90°). (c) It keeps growing, and it never reaches a largest value — as the run shrinks toward 0 the quotient $\frac{\text{rise}}{\text{run}}$ grows without bound. (d) At the end the segment is vertical, so the run is 0 and the slope is undefined. Infinity is not a number, so it cannot be the value of $\frac{\text{rise}}{\text{run}}$; the slope does not "reach" anything — it simply stops existing the instant the run becomes 0.

Lesson 8 — Extra Practice

Determining Collinearity Using Slope

Part A — The Slope Test, Including Some Very Near Misses

In every question below, show both slopes as exact fractions in lowest terms before you decide. Several of these sets *almost* line up — a sketch would not settle them, so the arithmetic has to.

1. Determine whether each set of points is collinear.
(a) $A(-5,-7)$, $B(-1,-1)$, $C(7,11)$ (b) $D(-6,4)$, $E(-2,1)$, $F(6,-6)$

2. Determine whether each set of points is collinear.
(a) $P(-8,-5)$, $Q(-2,-1)$, $R(7,5)$ (b) $S(-7,6)$, $T(-1,2)$, $U(8,-5)$

3. The line through $A(-6,-4)$ and $B(6,4)$ is drawn on a map. Decide whether each point below lies on that line. Justify each answer with a slope.
(a) $(-3,-2)$ (b) $(0,1)$ (c) $(3,2)$ (d) $(9,5)$

4. **Four points.** Consider $A(-6,-5)$, $B(-2,-2)$, $C(2,1)$, and $D(7,5)$.
(a) Are all four points collinear? Show the slopes you used.
(b) Exactly one set of *three* of these points is collinear. Which three, and how do you know the other three-point sets fail?

5. **Four more points.** Consider $W(-9,7)$, $X(-3,3)$, $Y(3,-1)$, and $Z(9,-5)$.
(a) Are all four collinear? Show enough slopes to be certain.
(b) There are four different three-point sets you could form from W, X, Y, Z. Explain why you do *not* need to test all four separately.

Part B — Finding the Missing Coordinate

When you are *told* the points are collinear, the two slopes are already equal. Set them equal and solve for the unknown. Set up the equation carefully — the unknown can sit on top or underneath.

6. Find the value of k that makes each set collinear.

(a) $A(-3,-4)$, $B(1,2)$, $C(5,k)$

(b) $D(-2,7)$, $E(2,1)$, $F(k,-5)$ — note the unknown is now an x-coordinate.

7. Find k. In each case the unknown point sits *between* the two known ones, so find the slope of the outer pair first.

(a) $P(-4,1)$, $Q(3,k)$, $R(10,9)$

(b) $S(-5,-2)$, $T(k,4)$, $U(7,10)$

(c) $E(-2,3)$, $F(4,k)$, $G(9,-4)$ — the answer is not an integer.

8. Harder: in each set the unknown appears **twice**.

(a) $A(0,1)$, $B(k,k)$, and $C(4,9)$ are collinear. Find k, then state the coordinates of B.

(b) $(-2,5)$, $(1,k)$, and $(7,k+8)$ are collinear. Find k. *Hint:* the slope between the last two points can be found without knowing k.

Part C — Two Different Methods, and Two Tempting Errors

There is a second way to test three points. If B really lies on segment AC, then travelling $A \to B \to C$ covers exactly the same distance as going straight from A to C, so $AB + BC = AC$. Distances come from $d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$.

9. Consider $A(-4,-1)$, $B(0,2)$, and $C(8,8)$.

(a) Test for collinearity using **slopes**.

(b) Test again using **distances**: find AB, BC, and AC, then check whether $AB + BC = AC$.

(c) Both methods agree here. Which was less work, and why?

10. Now consider $D(-3,-2)$, $E(1,1)$, and $F(9,8)$.

(a) Find DE, EF, and DF, each rounded to one decimal place. Does $DE + EF$ appear to equal DF?

(b) Now use slopes. What is the verdict?

(c) The two methods seem to disagree. Explain which one you trust and why rounding caused the trouble.

11. **Error analysis.** Marcus is asked whether $A(0,0)$, $B(2,3)$, $C(1,4)$, and $D(3,7)$ are collinear. He calculates $m_{AB} = \frac{3}{2}$ and $m_{CD} = \frac{3}{2}$, sees that they match, and answers "yes, all four are collinear."

(a) Both of his slopes are correct. Explain why his conclusion is still wrong. What do two equal slopes with *no shared point* actually tell you?

(b) Calculate m_{AC} to settle the question, and state the correct answer.

12. **Error analysis.** Testing $A(-2,3)$, $B(2,-1)$, $C(6,-5)$, Jaya writes

$$m_{AB} = \frac{y_A - y_B}{x_B - x_A} = \frac{3-(-1)}{2-(-2)} = 1, \qquad m_{BC} = \frac{y_C - y_B}{x_C - x_B} = \frac{-5-(-1)}{6-2} = -1,$$

and concludes the points are not collinear.

(a) Look closely at her first fraction. What did she do differently there, and why does it flip the sign?

(b) Redo m_{AB} correctly and give the right conclusion.

Part D — Data, Reasoning, Figures, and Stretch

13. A student hangs masses from a spring and records its length.

Mass (g)	Length (cm)
20	12.5
50	17.0
80	21.5

(a) Treating each row as a point $(\text{mass}, \text{length})$, decide whether the three points are collinear.

(b) Interpret the slope you found, with units. Then predict the spring's length at a mass of 100 g.

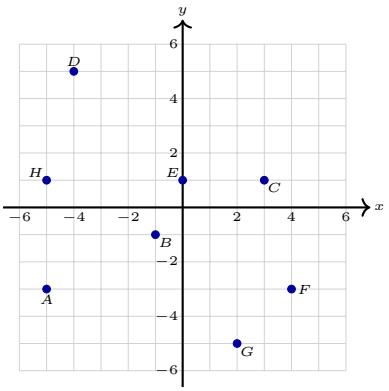
(c) A repeat of the experiment gives 21.5 cm replaced by 22.0 cm at 80 g. Are the new three points collinear? What would that suggest about the experiment?

14. **Reasoning.**

(a) Explain why asking "are these *two* points collinear?" is never an interesting question. What is always true of any two points?

(b) If A, B, and C are collinear, must B lie *between* A and C on the line? Give three specific collinear points where the middle-listed one is not in the middle.

15. Eight points are plotted below. Working from the grid coordinates, find **every** set of three of these points that is collinear, and give the slope of each such line. There are more than one, and one point belongs to no collinear triple at all.



Coordinates: $A(-5,-3)$, $B(-1,-1)$, $C(3,1)$, $D(-4,5)$, $E(0,1)$, $F(4,-3)$, $G(2,-5)$, $H(-5,1)$.
List each collinear triple with its slope, and name the leftover point.

16. **Stretch.** Triangle PQR has vertices $P(-4,-2)$, $Q(2,6)$, and $R(6,-2)$.

(a) Find M, the midpoint of PQ, and N, the midpoint of QR. Then show that M, N, and the vertex R are **not** collinear.

(b) Compare the slope of MN with the slope of PR. What do you notice, and why does that guarantee M, N, R could never have been collinear?

(c) Find **all** values of k for which $(0,0)$, $(k,4)$, and $(9,k)$ are collinear. There is more than one answer — find every one, and check each by computing both slopes.

Answer Key. 1. (a) $m_{AB} = \frac{6}{4} = \frac{3}{2}$, $m_{BC} = \frac{12}{8} = \frac{3}{2}$ — **collinear**. (b) $m_{DE} = \frac{-3}{4} = -\frac{3}{4}$, $m_{EF} = \frac{-7}{8} = -\frac{7}{8}$ — **not collinear** (close, but $-\frac{6}{8} \neq -\frac{7}{8}$). **2.** (a) $m_{PQ} = \frac{4}{6} = \frac{2}{3}$, $m_{QR} = \frac{6}{9} = \frac{2}{3}$ — **collinear**. (b) $m_{ST} = \frac{-4}{6} = -\frac{2}{3}$, $m_{TU} = \frac{-7}{9}$ — **not collinear** ($-\frac{6}{9} \neq -\frac{7}{9}$). **3.** $m_{AB} = \frac{8}{12} = \frac{2}{3}$. (a) yes, $m = \frac{2}{3}$ from A (b) no, $m_{A(0,1)} = \frac{5}{6}$ (c) yes, $m = \frac{6}{9} = \frac{2}{3}$ (d) no, $m_{A(9,5)} = \frac{9}{15} = \frac{3}{5}$. **4.** (a) No. $m_{AB} = \frac{3}{4}$, $m_{BC} = \frac{3}{4}$, but $m_{CD} = \frac{4}{5}$. (b) A, B, C only. $m_{AD} = \frac{10}{13}$, $m_{BD} = \frac{7}{9}$, $m_{CD} = \frac{4}{5}$ — every set containing D brings in a slope that does not equal $\frac{3}{4}$. **5.** (a) Yes: $m_{WX} = \frac{-4}{6} = -\frac{2}{3}$, $m_{XY} = \frac{-4}{6} = -\frac{2}{3}$, $m_{YZ} = \frac{-4}{6} = -\frac{2}{3}$. (b) Once W, X, Y are known to lie on one line and Y, Z have that same slope, Z is on that one line too — so all four are, and any three of them must be. Testing consecutive pairs is enough. **6.** (a) $m_{AB} = \frac{6}{4} = \frac{3}{2}$; $\frac{k-2}{4} = \frac{3}{2}$ gives $k = 8$. (b) $m_{DE} = \frac{-6}{4} = -\frac{3}{2}$; $\frac{-6}{k-2} = -\frac{3}{2}$ gives $k-2 = 4$, so $k = 6$. **7.** (a) $m_{PR} = \frac{8}{14} = \frac{4}{7}$; $\frac{k-1}{7} = \frac{4}{7}$ gives $k = 5$. (b) $m_{SU} = \frac{12}{12} = 1$; $\frac{6}{k+5} = 1$ gives $k = 1$. (c) $m_{EG} = \frac{-7}{11}$; $\frac{k-3}{6} = -\frac{7}{11}$ gives $k = 3 - \frac{42}{11} = -\frac{9}{11}$. **8.** (a) $m_{AC} = \frac{8}{4} = 2$, so $\frac{k-1}{k} = 2 \Rightarrow k-1 = 2k \Rightarrow k = -1$; $B(-1,-1)$, and indeed $m_{AB} = \frac{-2}{-1} = 2$. (b) The last two points give $m = \frac{(k+8)-k}{7-1} = \frac{8}{6} = \frac{4}{3}$ regardless of k. Then $\frac{k-5}{1-(-2)} = \frac{4}{3} \Rightarrow k-5 = 4 \Rightarrow k = 9$. **9.** (a) $m_{AB} = \frac{3}{4}$, $m_{BC} = \frac{6}{8} = \frac{3}{4}$ — collinear. (b) $AB = \sqrt{4^2+3^2} = 5$, $BC = \sqrt{8^2+6^2} = 10$, $AC = \sqrt{12^2+9^2} = \sqrt{225} = 15$, and $5+10 = 15$ — collinear. (c) The slope test: two subtractions and a division each, with exact fractions, versus three square roots. **10.** (a) $DE = 5.0$, $EF = \sqrt{113} \approx 10.6$, $DF = \sqrt{244} \approx 15.6$; $DE + EF \approx 15.6$, which appears to equal DF. (b) $m_{DE} = \frac{3}{4}$, $m_{EF} = \frac{7}{8}$, and $\frac{3}{4} \neq \frac{7}{8}$ — **not collinear**. (c) Trust the slopes. The distance sums differ by only about 0.01 (15.630 vs. 15.620), small enough to vanish under rounding, while the exact fractions $\frac{3}{4}$ and $\frac{7}{8}$ are plainly different. Exact comparisons beat rounded decimals. **11.** (a) Equal slopes with no point in common mean the two segments are *parallel*, not that they are the same line — AB and CD here are two different parallel segments. To use the slope test he must share a point, e.g. compare m_{AB} with m_{BC}. (b) $m_{AC} = \frac{4-0}{1-0} = 4 \neq \frac{3}{2}$, so the four points are **not** collinear. (Check: $m_{BC} = \frac{4-3}{1-2} = -1$ as well.) **12.** (a) In the first fraction she subtracted the y's in one order ($y_A - y_B$) and the x's in the opposite order ($x_B - x_A$). Reversing only one of the two subtractions multiplies the whole fraction by -1, so the sign flips. (b) $m_{AB} = \frac{-1-3}{2-(-2)} = \frac{-4}{4} = -1$, which equals $m_{BC} = -1$ — the points **are** collinear. **13.** (a) $\frac{17.0-12.5}{50-20} = \frac{4.5}{30} = 0.15$ and $\frac{21.5-17.0}{80-50} = \frac{4.5}{30} = 0.15$ — collinear. (b) The spring stretches 0.15 cm for each additional gram (cm per g). At 100 g: $21.5 + 0.15(20) = 24.5$ cm. (c) Now $\frac{22.0-17.0}{30} = \frac{5}{30} = \frac{1}{6} \approx 0.167 \neq 0.15$ — not collinear, so the second run is not perfectly linear; likely measurement error, or the spring was stretched past the range where it behaves linearly. **14.** (a) Any two distinct points always lie on exactly one line, so two points are always collinear — there is nothing to decide. Collinearity only becomes a real question with three or more points. (b) No. Listing order has nothing to do with position. For example $(0,0)$, $(6,6)$, $(2,2)$ are collinear (all slope 1), yet the

middle-listed point $(6,6)$ is the far end, and $(2,2)$ is the one actually in between. **15.** Three triples: A, B, C with slope $\frac{2}{4}=\frac{1}{2}$; D, E, F with slope $\frac{-4}{4}=-1$; H, E, C with slope 0 (all three have $y=1$, a horizontal line). The leftover point is $G(2,-5)$, which lies on none of them. **16.** (a) $M=\left(\frac{-4+2}{2},\frac{-2+6}{2}\right)=(-1,2)$ and $N=\left(\frac{2+6}{2},\frac{6+(-2)}{2}\right)=(4,2)$. Then $m_{MN}=\frac{2-2}{4-(-1)}=0$ but $m_{NR}=\frac{-2-2}{6-4}=-2$; since $0\neq -2$, M, N, R are **not** collinear. (b) $m_{PR}=\frac{-2-(-2)}{6-(-4)}=0=m_{MN}$, so MN is parallel to PR. A line through M and N is parallel to PR and does not contain R, so R can never fall on it — the midpoint segment MN always misses the opposite vertex. (c) The outer slope condition $\frac{4-0}{k-0}=\frac{k-0}{9-0}$ gives $k^2=36$, so $k=6$ **or** $k=-6$. Check $k=6$: $(0,0)$, $(6,4)$, $(9,6)$ have slopes $\frac{4}{6}=\frac{2}{3}$ and $\frac{2}{3}$. Check $k=-6$: $(0,0)$, $(-6,4)$, $(9,-6)$ have slopes $\frac{4}{-6}=-\frac{2}{3}$ and $\frac{-10}{15}=-\frac{2}{3}$. Both work.

Lesson 9 — Extra Practice

Parallel and Perpendicular Lines

Part A — Deciding from Two Pairs of Points

For every pair below, calculate both slopes as exact fractions in lowest terms, then state whether the segments are **parallel**, **perpendicular**, or **neither**. Several pairs sit very close to perpendicular without actually being perpendicular, so the check $m_1 \times m_2 = -1$ has to be carried out — a sketch will not settle them.

1. Calculate both slopes, multiply them, and state the relationship.

(a) $A(-7,-4)$, $B(5,2)$ and $C(-3,6)$, $D(1,-2)$

(b) $E(-6,5)$, $F(2,-1)$ and $G(-4,-6)$, $H(4,-12)$

(c) $J(-5,-2)$, $K(3,4)$ and $L(-2,7)$, $M(5,-1)$

2. Calculate both slopes, multiply them, and state the relationship.

(a) $P(-8,3)$, $Q(-2,-1)$ and $R(1,-5)$, $S(7,-9)$

(b) $T(-6,-5)$, $U(2,1)$ and $V(-1,4)$, $W(7,-2)$

(c) $X(-4,6)$, $Y(2,-4)$ and $Z(-5,-3)$, $N(5,3)$

3. These pairs include horizontal and vertical segments. Decide the relationship, and for each one say whether the rule $m_1 \times m_2 = -1$ could be used or not.

(a) $A(-6,3)$, $B(4,3)$ and $C(2,-5)$, $D(2,6)$

(b) $E(-3,-4)$, $F(-3,5)$ and $G(6,-2)$, $H(6,7)$

(c) $J(-5,2)$, $K(3,2)$ and $L(-4,-6)$, $M(6,-1)$

Part B — Negative Reciprocals, Including the Two Special Cases

"Flip it and change the sign" works for every ordinary slope. It does *not* work on a slope of 0, because $\frac{0}{1}$ flipped is $\frac{1}{0}$, which is not a number. Parts (g) and (h) of question 4 are where the rule runs out and you have to think about the picture instead.

4. For each slope m, state the slope of a line **perpendicular** to it and the slope of a line **parallel** to it.

(a) $m = \frac{2}{3}$ (b) $m = -4$ (c) $m = 1$ (d) $m = -\frac{1}{6}$

(e) $m = \frac{7}{5}$ (f) $m = -\frac{9}{2}$ (g) $m = 0$ (h) m is undefined

5. Look again at parts (g) and (h) above.

(a) Explain, using a picture of a horizontal line and a vertical line, why they must be perpendicular even though you cannot multiply their slopes to get -1.

(b) A student says "a slope of 0 and an undefined slope are the same thing — both mean there is no slope." Explain what is wrong with this, and describe the line in each case.

6. The two slopes given describe **parallel** lines. Solve for the variable.

(a) $\frac{5}{6}$ and $\frac{k}{18}$ (b) $-\frac{7}{4}$ and $-\frac{21}{w}$ (c) -3 and $\frac{n}{4}$ (d) $\frac{8}{k}$ and $\frac{2}{5}$

7. The two slopes given describe **perpendicular** lines. Solve for the variable.

(a) $\frac{3}{8}$ and $-\frac{16}{k}$ (b) $-\frac{2}{5}$ and $\frac{t}{2}$ (c) 6 and $\frac{w}{12}$ (d) $-\frac{5}{9}$ and $\frac{45}{n}$

8. **Stretch.** Line ℓ_1 has slope $\dfrac{k}{4}$ and line ℓ_2 has slope $\dfrac{9}{k}$, where $k \neq 0$.

(a) Find every value of k for which $\ell_1 \parallel \ell_2$. There is more than one answer.

(b) Show that no value of k makes $\ell_1 \perp \ell_2$. *Hint:* simplify the product of the two slopes before you solve anything.

Part C — Finding a Missing Coordinate

When you are told two segments are parallel, their slopes are already equal; when you are told they are perpendicular, one slope is the negative reciprocal of the other. Either way you get an equation in the unknown coordinate. Find the slope of the *complete* segment first.

9. Segment AB joins $A(-3,-2)$ and $B(9,k)$. Segment CD joins $C(-1,4)$ and $D(7,-2)$.

(a) Find k so that $AB \parallel CD$.

(b) Find k so that $AB \perp CD$.

(c) The run of AB was 12 in both parts. Explain why the two values of k are so far apart.

10. Now the unknown is an x-coordinate. Segment PQ joins $P(-4, 5)$ and $Q(k, -3)$. Segment RS joins $R(-6, 3)$ and $S(-2, -5)$.

(a) Find k so that $PQ \parallel RS$.

(b) Find k so that $PQ \perp RS$.

Part D — Proving a Figure Is What It Claims to Be

A figure is a rectangle only if *both* conditions hold: the two pairs of opposite sides are parallel, *and* one pair of adjacent sides is perpendicular. Checking only the first condition proves a parallelogram and nothing more.

11. Quadrilateral $ABCD$ has vertices $A(-5, -3)$, $B(3, 3)$, $C(0, 7)$, and $D(-8, 1)$, joined in that order.

(a) Find the slope of each of the four sides AB, BC, CD, and DA.

(b) Use your slopes to show that $AB \parallel CD$ and that $BC \parallel DA$.

(c) Use your slopes to show that $AB \perp BC$ and that $CD \perp DA$.

(d) Find the length of each side. Is $ABCD$ a square? Explain.

(e) State the most precise name for $ABCD$ and give the perimeter.

(f) Which single one of parts (b) and (c) would you have to drop to leave yourself with only a parallelogram?

12. Triangle PQR has vertices $P(-6, -2)$, $Q(0, 2)$, and $R(-8, 1)$.

(a) Use **slopes only** to show the triangle has a right angle, and say at which vertex.

(b) Now use the **distance formula and the Pythagorean theorem** to confirm the same result.

(c) Which method took less work here, and why? When would the other method be the better choice?

13. **Stretch.** Three vertices of a rectangle are $A(-5,-1)$, $B(1,1)$, and $C(0,4)$, joined in that order.

(a) Verify that $AB \perp BC$, so a right angle really does sit at B.

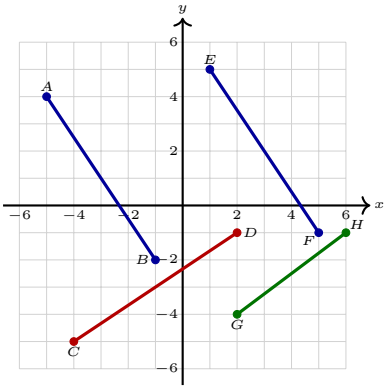
(b) Find the coordinates of the fourth vertex D, then verify that $CD \parallel AB$ and $DA \parallel BC$.

(c) Find all four side lengths and decide whether the rectangle is a square.

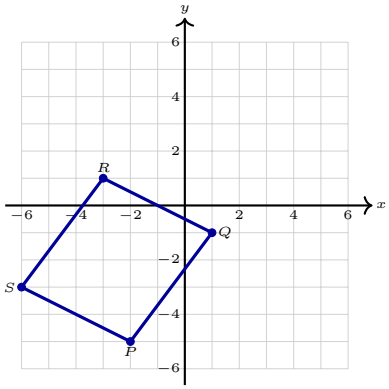
Part E — Reading the Relationships off a Diagram

14. Read the endpoints off each grid, calculate every slope you need, and answer in full.

(a) List every pair of the four segments that is parallel, and every pair that is perpendicular. Some segments belong to no pair at all.



(b) Quadrilateral $PQRS$ below looks like a rectangle. Find all four slopes and decide whether it really is one. Name the figure as precisely as you can.



Part F — Error Analysis, Reasoning, and an Application

15. Dmitri is asked for the slope of a line perpendicular to a line of slope $\frac{5}{8}$. He writes $\frac{8}{5}$, saying "perpendicular means flip the fraction." Multiply his two slopes to show he is wrong, state the correct answer, and describe the mistake in one sentence.

16. Priya claims the lines with slopes 2 and -2 are perpendicular "because one is the negative of the other." Multiply the slopes to show she is wrong, give the slope that really is perpendicular to 2, and describe what the relationship between 2 and -2 actually is.

17. Can two lines that both have *positive* slope ever be perpendicular? Answer yes or no, justify it using the product $m_1 \times m_2$, and then explain the same thing a second way by describing how the two lines would look on a grid.

18. A student writes the rule "if two lines are perpendicular, one slope is positive and the other is negative." The rule is *almost* right. Find the one pair of perpendicular lines in this unit for which it fails, and restate the rule so that it is always true.

19. A town's street grid is drawn on coordinate axes. Elm Street runs through $(-8, 5)$ and $(4, -1)$; Birch Street runs through $(-3, -6)$ and $(3, 6)$; Cedar Street runs through $(0, 7)$ and $(10, 2)$.

(a) Find the slope of each of the three streets.

(b) State every pair of streets that meets at a right angle, and every pair that never meets.

(c) A fourth street must be parallel to Birch Street. What slope must it have, and could it ever cross Elm Street?

Answer Key. 1. (a) $m_{AB} = \frac{1}{2}$, $m_{CD} = -2$, product -1: **perpendicular**. (b) both $-\frac{3}{4}$: **parallel**. (c) $m_{JK} = \frac{3}{4}$, $m_{LM} = -\frac{8}{7}$, product $-\frac{6}{7} \neq -1$: **neither** — the negative reciprocal of $\frac{3}{4}$ is $-\frac{4}{3}$, and $-\frac{8}{7}$ is close to that but not equal to it. **2.** (a) both $-\frac{2}{3}$: parallel. (b) $\frac{3}{4}$ and $-\frac{3}{4}$, product $-\frac{9}{16}$: neither (opposite signs alone never make a right angle). (c) $-\frac{5}{3}$ and $\frac{3}{5}$, product -1: perpendicular. **3.** (a) 0 and undefined: perpendicular (horizontal meets vertical), but the product rule cannot be used — an undefined slope is not a number to multiply by. (b) both undefined: parallel (two vertical lines); again no product. (c) 0 and $\frac{1}{2}$: neither; the product 0 is a legal calculation here, and $0 \neq -1$. **4.** perpendicular / parallel: (a) $-\frac{3}{2}$ / $\frac{2}{3}$ (b) $\frac{1}{4}$ / -4 (c) -1 / 1 (d) 6 / $-\frac{1}{6}$ (e) $-\frac{5}{7}$ / $\frac{7}{5}$ (f) $\frac{2}{9}$ / $-\frac{9}{2}$ (g) **undefined** (a vertical line) / 0 (h) 0 (a horizontal line) / undefined. **5.** (a) A horizontal line runs flat and a vertical line runs straight up and down, so they cross at exactly 90°; "flip it" turns $\frac{0}{1}$ into $\frac{1}{0}$, which is division by zero, so the algebra breaks even though the geometry does not. (b) They are opposites, not the same. Slope 0 is a genuine number and describes a flat horizontal line ($y = c$); an undefined slope is not a number at all and describes a vertical line ($x = c$). **6.** (a) $k = 15$ (b) $w = 12$ (c) $n = -12$ (d) $k = 20$. **7.** (a) $k = 6$ (b) $t = 5$ (c) $w = -2$ (d) $n = 25$. **8.** (a) $\frac{k}{4} = \frac{9}{k} \Rightarrow k^2 = 36 \Rightarrow k = 6$ or $k = -6$. (b) $\frac{k}{4} \times \frac{9}{k} = \frac{9}{4}$ for every $k \neq 0$ — the k's cancel — and $\frac{9}{4}$ is never -1, so no value of k works. **9.** $m_{CD} = -\frac{3}{4}$. (a) $\frac{k+2}{12} = -\frac{3}{4} \Rightarrow k = -11$. (b) $\frac{k+2}{12} = \frac{4}{3} \Rightarrow k = 14$. (c) Over the same run of 12, the parallel case falls 9 while the perpendicular case rises 16 — opposite directions and different steepness, a gap of 25 in k. **10.** $m_{RS} = -2$. (a) $\frac{-8}{k+4} = -2 \Rightarrow k = 0$. (b) $\frac{-8}{k+4} = \frac{1}{2} \Rightarrow k = -20$. **11.** (a) $m_{AB} = \frac{3}{4}$, $m_{BC} = -\frac{4}{3}$, $m_{CD} = \frac{3}{4}$, $m_{DA} = -\frac{4}{3}$. (b) $m_{AB} = m_{CD}$ and $m_{BC} = m_{DA}$, so both pairs of opposite sides are parallel. (c) $\frac{3}{4} \times \left(-\frac{4}{3}\right) = -1$ for AB and BC, and likewise for CD and DA. (d) $AB = CD = 10$, $BC = DA = 5$; not a square, since adjacent sides are unequal. (e) A **rectangle**; perimeter $= 2(10) + 2(5) = 30$. (f) Drop part (c): without a right angle the parallel pairs alone give only a parallelogram. **12.** (a) $m_{PQ} = \frac{2}{3}$, $m_{PR} = -\frac{3}{2}$, product -1, so the right angle is at P. (b) $PQ = \sqrt{52} = 2\sqrt{13}$, $PR = \sqrt{13}$, $QR = \sqrt{65}$; $52 + 13 = 65$, so $PQ^2 + PR^2 = QR^2$ and the triangle is right-angled with QR as hypotenuse. (c) Slopes were faster — two subtractions each and one multiplication, with no radicals. Distances are the better choice when you also need the side lengths (for perimeter, or to classify the triangle by its sides), or when the right angle is not at a labelled vertex you can test directly. **13.** (a) $m_{AB} = \frac{1}{3}$, $m_{BC} = -3$, product -1. (b) $D(-6, 2)$; $m_{CD} = \frac{1}{3} = m_{AB}$ and $m_{DA} = -3 = m_{BC}$. (c) $AB = CD = \sqrt{40} = 2\sqrt{10}$, $BC = DA = \sqrt{10}$; not a square, because $2\sqrt{10} \neq \sqrt{10}$. **14.** (a) $m_{AB} = -\frac{3}{2}$, $m_{CD} = \frac{2}{3}$, $m_{EF} = -\frac{3}{2}$, $m_{GH} = \frac{3}{4}$. Parallel: $AB \parallel EF$. Perpendicular: $AB \perp CD$ and $EF \perp CD$ (each product is -1). GH is neither parallel nor perpendicular to any of them: with AB and with EF the product is $-\frac{9}{8}$, and with CD it is $\frac{1}{2}$. (b) $m_{PQ} = \frac{4}{3}$, $m_{QR} = -\frac{1}{2}$, $m_{RS} = \frac{4}{3}$, $m_{SP} = -\frac{1}{2}$. Opposite sides are parallel, but $\frac{4}{3} \times \left(-\frac{1}{2}\right) = -\frac{2}{3} \neq -1$, so no corner is square: it is a **parallelogram**, not a rectangle. **15.** $\frac{5}{8} \times \frac{8}{5} = 1$, not -1. The correct slope is $-\frac{8}{5}$. He flipped the fraction but forgot to change the sign — the reciprocal alone gives a product of $+1$. **16.** $2 \times (-2) = -4 \neq -1$. The slope perpendicular to 2 is $-\frac{1}{2}$. Slopes 2 and -2 are mirror images: equally steep, one rising and one falling, like the two sides of a roof — steepness is the same, but the angle between them is not 90°. **17.** No. If $m_1 > 0$ and $m_2 > 0$ then $m_1 \times m_2 > 0$, and a positive number can never equal -1. Geometrically, both lines rise as you move right; to turn one of them a quarter-turn you must tip it past vertical so it starts falling, which makes that slope negative. **18.** It fails for a horizontal line and a vertical line: slope 0 is neither positive nor negative, and the other slope is undefined, yet the lines are perpendicular. Corrected rule: "if two perpendicular lines both have defined, non-zero slopes, one is positive and the other is negative; the only other perpendicular pair is a horizontal line with a vertical line." **19.** (a) Elm $-\frac{1}{2}$, Birch 2, Cedar $-\frac{1}{2}$. (b) Right angles: Elm $\perp$ Birch and Cedar $\perp$ Birch (each product is -1). Never meet: Elm $\parallel$ Cedar (equal slopes, different streets). (c) Slope 2; yes — it is parallel to Birch, and Birch itself crosses Elm, so any line of slope 2 crosses every line of slope $-\frac{1}{2}$ exactly once.

Lesson 10 — Extra Practice

Unit Review: Coordinate Geometry

Part A — Lengths and Midpoints

Nothing in this set uses a single formula on its own. Every question needs at least two of the nine ideas from the unit, and most need three or four. Keep all lengths exact (simplified mixed radicals), keep all slopes as fractions in lowest terms, and label which formula you are using at each step.

1. **Working backwards.** The midpoint of segment AB is $M(2, 3)$. The slope of AB is $\frac{3}{4}$, and the x-coordinate of A is -6.

(a) Use the midpoint formula to find the x-coordinate of B, and state the run of AB.

(b) Use the slope to find the rise, then find the two y-coordinates. *Hint:* they must average to 3.

(c) State A and B, then find the exact length of AB.

2. $M(2, -1)$ is the midpoint of segment AB, and A is the point $(-4, 3)$.

(a) Find the coordinates of B and the slope of AB.

(b) Find the exact lengths of AB and of AM.

(c) Show that AM is exactly half of AB, working with the radicals rather than decimals.

3. **The perpendicular bisector.** Segment AB joins $A(-5, -2)$ and $B(3, 4)$.

(a) Find the midpoint M of AB and the slope of AB.

(b) State the slope of the line through M that is perpendicular to AB, then show that $P(2, -3)$ lies on that line.

(c) Use the distance formula to show P is the same distance from A as it is from B.

Part B — Slope and Direction

4. The points $P(-9,-5)$, $Q(-3,-1)$, and $R(6,5)$ are plotted on a grid.

(a) Use slopes to show the three points are collinear.

(b) Find the exact lengths PQ, QR, and PR.

(c) Show that $PQ + QR = PR$, and explain why that equation is a second proof of collinearity.

(d) Is Q the midpoint of PR? Use the midpoint formula, then explain your answer using the lengths in (b).

5. The points $A(-8,-7)$, $B(-2,-3)$, and $C(k,3)$ are collinear.

(a) Find the value of k.

(b) Using your value of k, find the exact length of AC.

(c) Find the midpoint of AC and verify with a slope that it also lies on the same line.

6. Segment AB joins $A(-4,6)$ and $B(5,6)$; segment CD joins $C(3,-2)$ and $D(3,7)$; segment EF joins $E(-6,-5)$ and $F(-1,-5)$.

(a) State the slope of each segment and classify it as positive, negative, zero, or undefined.

(b) Find the length of each segment without using the distance formula, and say why you are allowed to.

(c) State which pairs are parallel and which are perpendicular, and explain why the rule $m_1 \times m_2 = -1$ cannot be used here.

7. Segment GH joins $G(-5,1)$ and $H(k,7)$. Segment JK joins $J(-2,-4)$ and $K(4,-8)$.

(a) Find k so that $GH \parallel JK$.

(b) Find k so that $GH \perp JK$.

(c) Using the value of k from (b), find the exact length of GH.

Part C — Classifying Figures

Questions 8 and 9 use the two grids below. Read every coordinate off the grid before you start, and quote a slope or a length as the reason for each claim you make.

Figure 1 — quadrilateral $ABCD$ (question 8)

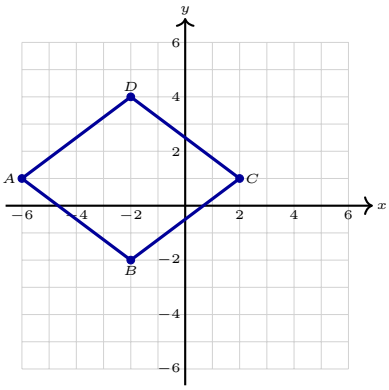
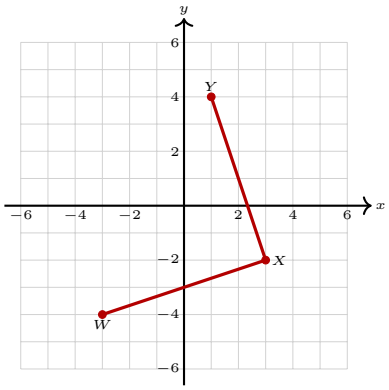


Figure 2 — three vertices W, X, Y (question 9)



8. Quadrilateral $ABCD$ in Figure 1 has vertices $A(-6, 1)$, $B(-2, -2)$, $C(2, 1)$, and $D(-2, 4)$.

(a) Find the slope of each of the four sides.

(b) Find the length of each of the four sides, and the perimeter.

(c) Which pairs of sides are parallel? Are any two adjacent sides perpendicular? Show the products.

(d) Find the midpoint of diagonal AC and the midpoint of diagonal BD. Do the diagonals bisect each other?

(e) Find the slope and the length of each diagonal. Are the diagonals perpendicular? Are they equal?

(f) Give the most precise name for $ABCD$, and state exactly which of your results rules out "square".

9. In Figure 2, $W(-3, -4)$, $X(3, -2)$, and $Y(1, 4)$ are three vertices of a square $WXYZ$, named in order.

(a) Show that $WX \perp XY$ and that $WX = XY$.

(b) Find the fourth vertex Z, and verify it with two slopes.

(c) Find the exact length of each diagonal, and the midpoint of each.

(d) List the three separate facts that together prove $WXYZ$ is a square.

10. Quadrilateral $EFGH$ has vertices $E(-7,-4)$, $F(-3,2)$, $G(5,4)$, and $H(1,-2)$.

(a) Find the slope and the exact length of each of the four sides.

(b) Find the midpoint and the exact length of each diagonal.

(c) Give the most precise name for $EFGH$, with reasons.

(d) Tomas says "the diagonals bisect each other, so it is a rectangle." Explain his error using two of your own results.

11. Triangle ABC has vertices $A(-6,-4)$, $B(0,-8)$, and $C(2,8)$.

(a) Find the midpoint of each of the three sides.

(b) Find the exact length of each side, then classify the triangle as scalene, isosceles, or equilateral.

(c) Classify the triangle by its angles, twice: once with slopes, and once with the converse of the Pythagorean theorem.

(d) Find the exact length of the median from A to the midpoint of BC (a **median** joins a vertex to the midpoint of the opposite side). Compare it with half of BC and comment.

Part D — Reasoning and Error Analysis

12. For the points $A(-5,3)$ and $B(7,-2)$, a student submitted the work below. Three separate formulas are used, and each one is wrong in a different way.

$$d = \sqrt{\big(7-(-5)\big)^2 - \big(-2-3\big)^2} = \sqrt{144-25} = \sqrt{119} \qquad M = \left(\frac{7-(-5)}{2},\ \frac{-2-3}{2}\right) = (6,-2.5) \qquad m = \frac{7-(-5)}{-2-3} = -\frac{12}{5}$$

(a) Name the mistake in the distance calculation and redo it. The correct answer is a whole number.

(b) Name the mistake in the midpoint calculation and redo it.

(c) Name the mistake in the slope calculation and redo it. Which two of these three errors produce answers that are not even close to correct?

13. State whether each claim is true or false. If it is false, give a specific counterexample with coordinates; if it is true, explain why in one sentence.

(a) Two segments with the same length must have the same slope.

(b) Two different segments can have the same midpoint.

(c) If $m_{AB} = m_{BC}$, then A, B, and C are collinear.

(d) A segment whose slope is undefined has length $|y_2 - y_1|$.

(e) If two slopes have opposite signs, the lines are perpendicular.

(f) A quadrilateral with four equal sides must be a square.

14. **Stretch.** Name the most precise quadrilateral that must result from each set of diagonal properties. Assume in every case that the figure is a quadrilateral whose diagonals do the following.

(a) bisect each other

(b) bisect each other and are equal in length

(c) bisect each other and are perpendicular

(d) bisect each other, are equal, and are perpendicular

Part E — Applications

15. A town map is drawn on coordinate axes with one grid unit equal to 100 m. The school is at $S(-8, -4)$, the library is at $L(4, 1)$, and the arena is at $A(9, -11)$.

(a) Find the distance from the school to the library, and from the library to the arena, in metres.

(b) Find the slope of SL and of LA. Does the route turn a square corner at the library?

(c) A bus shelter goes exactly halfway between the school and the arena. Give its coordinates.

(d) Find the direct distance from the school to the arena, to the nearest metre. How much farther is the route through the library?

16. A drone launches at $(-10, -6)$, passes a checkpoint at $(2, 3)$, and lands at $(10, 9)$ on a map where one unit is 50 m.

(a) Use slopes to decide whether the drone flew in a straight line the whole way.

(b) Find the distance flown in each stage and in total, in metres.

(c) Was the checkpoint halfway? Answer with the midpoint formula, then say what fraction of the trip it actually was.

17. A straight section of highway runs from $(-6,-2)$ to $(9,6)$ on a map where one unit is 1 km.

(a) Find the length of the highway and its slope.

(b) A service road leaves the highway at its midpoint, at right angles to it. Give the midpoint and the slope of the service road.

(c) A second road runs from $(-4,5)$ to $(4,-10)$. Is it parallel to the service road? Show the slopes.

Part F — Challenge

18. **Two answers.** Point A is $(6,-2)$.

(a) Find *both* points on the y-axis that are exactly 10 units from A. *Hint:* the point is $(0,y)$; substitute into the distance formula and solve.

(b) Find the slope of the segment from A to each of those two points. What do the two slopes have in common, and how do they differ?

(c) Are those two segments perpendicular to each other? Justify with the product of the slopes.

19. Find the point on the x-axis that is the same distance from $A(-6,3)$ as it is from $B(2,-5)$. Do it twice: once by setting two distance expressions equal, and once by finding the perpendicular bisector of AB and seeing where it crosses the x-axis.

20. Three vertices of a parallelogram are $A(-7,-3)$, $B(-1,1)$, and $C(5,-2)$. There are **three** different points D that complete a parallelogram — one for each choice of which pair of points forms a diagonal. Find all three, and state which pair of points is the diagonal in each case.

Answer Key. 1. (a) $x_B = 2(2) - (-6) = 10$; run $= 10 - (-6) = 16$. (b) rise $= \frac{3}{4}(16) = 12$, so $y_B - y_A = 12$ and $y_A + y_B = 6$, giving $y_A = -3$, $y_B = 9$. (c) $A(-6,-3)$, $B(10,9)$; $AB = \sqrt{16^2 + 12^2} = \sqrt{400} = 20$. **2.** (a) $B(2(2) - (-4), 2(-1) - 3) = B(8,-5)$; $m = \frac{-5-3}{8-(-4)} = -\frac{2}{3}$. (b) $AB = \sqrt{12^2 + 8^2} = \sqrt{208} = 4\sqrt{13}$; $AM = \sqrt{6^2 + 4^2} = \sqrt{52} = 2\sqrt{13}$. (c) $2 \times 2\sqrt{13} = 4\sqrt{13} = AB$. **3.** (a) $M(-1,1)$; $m_{AB} = \frac{6}{8} = \frac{3}{4}$. (b) Perpendicular slope $-\frac{4}{3}$; from $M(-1,1)$ to $P(2,-3)$ the slope is $\frac{-4}{3} = -\frac{4}{3}$, so P is on the line. (c) $PA = \sqrt{49+1} = \sqrt{50} = 5\sqrt{2}$ and $PB = \sqrt{1+49} = 5\sqrt{2}$; equal. **4.** (a) $m_{PQ} = \frac{4}{6} = \frac{2}{3}$ and $m_{QR} = \frac{6}{9} = \frac{2}{3}$; equal slopes through the shared point Q, so collinear. (b) $PQ = \sqrt{52} = 2\sqrt{13}$, $QR = \sqrt{117} = 3\sqrt{13}$, $PR = \sqrt{325} = 5\sqrt{13}$. (c) $2\sqrt{13} + 3\sqrt{13} = 5\sqrt{13}$. Going $P \to Q \to R$ covers exactly the direct distance, so no detour is made — Q must be on segment PR. (d) No. The midpoint of PR is $\left(-\frac{3}{2}, 0\right)$, not $Q(-3,-1)$; Q is only $2\sqrt{13}$ of the $5\sqrt{13}$ along, that is $\frac{2}{5}$ of the way. **5.** (a) $m_{AB} = \frac{4}{6} = \frac{2}{3}$, so $\frac{3-(-3)}{k-(-2)} = \frac{2}{3}$, giving $k + 2 = 9$ and $k = 7$. (b) $AC = \sqrt{15^2 + 10^2} = \sqrt{325} = 5\sqrt{13}$. (c) Midpoint $\left(-\frac{1}{2}, -2\right)$; slope from $A(-8,-7)$ to it is $\frac{5}{7.5} = \frac{2}{3}$, matching. **6.** (a) $m_{AB} = 0$ (zero), m_{CD} undefined, $m_{EF} = 0$ (zero). (b) $AB = |5 - (-4)| = 9$, $CD = |7 - (-2)| = 9$, $EF = |-1 - (-6)| = 5$; the segments are horizontal or vertical, so the run or the rise is 0 and the distance formula collapses to a single absolute value. (c) $AB \parallel EF$ (both horizontal); $AB \perp CD$ and $EF \perp CD$ (horizontal meets vertical). The product rule needs two numbers, and an undefined slope is not a number. **7.** $m_{JK} = \frac{-4}{6} = -\frac{2}{3}$. (a) $\frac{6}{k+5} = -\frac{2}{3} \Rightarrow k + 5 = -9 \Rightarrow k = -14$. (b) $\frac{6}{k+5} = \frac{3}{2} \Rightarrow k + 5 = 4 \Rightarrow k = -1$. (c) $H(-1,7)$, so $GH = \sqrt{4^2 + 6^2} = \sqrt{52} = 2\sqrt{13}$. **8.** (a) $m_{AB} = -\frac{3}{4}$, $m_{BC} = \frac{3}{4}$, $m_{CD} = -\frac{3}{4}$, $m_{DA} = \frac{3}{4}$. (b) All four sides are 5; perimeter 20. (c) $AB \parallel CD$ and $BC \parallel DA$. Adjacent: $-\frac{3}{4} \times \frac{3}{4} = -\frac{9}{16} \neq -1$, so no corner is a right angle. (d) Both diagonals have midpoint $(-2,1)$, so yes, they bisect each other. (e) AC is horizontal, slope 0, length 8; BD is vertical, slope undefined, length 6; they are perpendicular (horizontal with vertical) but not equal. (f) A **rhombus**: four equal sides, opposite sides parallel, perpendicular diagonals — but the corners are not right angles (part (c)) and the diagonals are unequal (part (e)), so it is not a square. **9.** (a) $m_{WX} = \frac{2}{6} = \frac{1}{3}$, $m_{XY} = \frac{6}{-2} = -3$, product -1; $WX = \sqrt{36+4} = 2\sqrt{10}$ and $XY = \sqrt{4+36} = 2\sqrt{10}$. (b) $Z(-5,2)$; $m_{YZ} = \frac{1}{3}$ matches WX and $m_{ZW} = -3$ matches XY. (c) $WY = \sqrt{16+64} = \sqrt{80} = 4\sqrt{5}$ and

$XZ = \sqrt{64+16} = 4\sqrt{5}$; both have midpoint $(-1, 0)$. (d) All four sides equal $2\sqrt{10}$; adjacent sides are perpendicular; the diagonals are equal and bisect each other. **10.** (a) $m_{EF} = m_{GH} = \frac{3}{2}$ with length $\sqrt{52} = 2\sqrt{13}$; $m_{FG} = m_{HE} = \frac{1}{4}$ with length $\sqrt{68} = 2\sqrt{17}$. (b) EG: midpoint $(-1, 0)$, length $\sqrt{208} = 4\sqrt{13}$; FH: midpoint $(-1, 0)$, length $\sqrt{32} = 4\sqrt{2}$. (c) A **parallelogram** only — both pairs of opposite sides are parallel and equal, but $\frac{3}{2} \times \frac{1}{4} = \frac{3}{8} \neq -1$ and adjacent sides are unequal. (d) Bisecting diagonals prove a parallelogram and nothing more. A rectangle also needs equal diagonals, and here $4\sqrt{13} \approx 14.4$ while $4\sqrt{2} \approx 5.7$; equivalently, no corner is a right angle. **11.** (a) AB: $(-3, -6)$; AC: $(-2, 2)$; BC: $(1, 0)$. (b) $AB = \sqrt{52} = 2\sqrt{13}$, $AC = \sqrt{208} = 4\sqrt{13}$, $BC = \sqrt{260} = 2\sqrt{65}$; all different, so **scalene**. (c) Slopes: $m_{AB} = -\frac{2}{3}$ and $m_{AC} = \frac{3}{2}$, product -1, so the right angle is at A. Pythagoras: $52 + 208 = 260$, so $AB^2 + AC^2 = BC^2$ and the triangle is right-angled with BC as hypotenuse. It is a scalene right triangle. (d) Median from $A(-6, -4)$ to $(1, 0)$: $\sqrt{49+16} = \sqrt{65}$. Half of BC is $\frac{2\sqrt{65}}{2} = \sqrt{65}$ — the same. In any right triangle the median from the right angle is exactly half the hypotenuse. **12.** (a) The two squares must be *added*, not subtracted: $d = \sqrt{144+25} = \sqrt{169} = 13$. (b) The midpoint formula *averages*, so the coordinates are added, not subtracted: $M\left(\frac{-5+7}{2}, \frac{3+(-2)}{2}\right) = (1, 0.5)$. (c) The slope was written upside down (run over rise): $m = \frac{-2-3}{7-(-5)} = -\frac{5}{12}$. The midpoint and the slope are badly wrong — $(6, -2.5)$ is not even between A and B, and $-\frac{12}{5}$ is nearly six times too steep — while $\sqrt{119} \approx 10.9$ at least lands in the right neighbourhood as 13, which is exactly why that error is the easiest to miss. **13.** (a) False — $(0, 0)$ to $(5, 0)$ and $(0, 0)$ to $(0, 5)$ both have length 5, with slopes 0 and undefined. (b) True — the two diagonals of any parallelogram share a midpoint, for example $(-3, 0)$ to $(3, 0)$ and $(0, -4)$ to $(0, 4)$, both centred at $(0, 0)$. (c) True — equal slopes through the shared point B means C sits on the same line as A and B. (d) True — an undefined slope means the run is 0, so the segment is vertical and its length is the change in y. (e) False — $\frac{3}{4}$ and $-\frac{1}{2}$ have opposite signs but their product is $-\frac{3}{8} \neq -1$. (f) False — the rhombus $A(-6, 1)$, $B(-2, -2)$, $C(2, 1)$, $D(-2, 4)$ of question 8 has four sides of length 5 and no right angle. **14.** (a) parallelogram (b) rectangle (c) rhombus (d) square. **15.** (a) $SL = \sqrt{12^2 + 5^2} = 13$ units = 1300 m; $LA = \sqrt{5^2 + 12^2} = 13$ units = 1300 m. (b) $m_{SL} = \frac{5}{12}$, $m_{LA} = -\frac{12}{5}$, product -1: yes, a square corner at the library. (c) $\left(\frac{1}{2}, -\frac{15}{2}\right)$, that is $(0.5, -7.5)$. (d) $SA = \sqrt{17^2 + 7^2} = \sqrt{338} = 13\sqrt{2} \approx 18.38$ units ≈ 1838 m; the route through the library is 2600 m, about 762 m farther. **16.** (a) $m = \frac{9}{12} = \frac{3}{4}$ for the first stage and $\frac{6}{8} = \frac{3}{4}$ for the second: equal slopes through the checkpoint, so the whole flight is straight. (b) $\sqrt{144+81} = 15$ units = 750 m, then $\sqrt{64+36} = 10$ units = 500 m; total 25 units = 1250 m, which matches the direct distance $\sqrt{400+225} = 25$ units. (c) No — the midpoint of launch and landing is $\left(0, \frac{3}{2}\right)$, not $(2, 3)$. The checkpoint is 15 of the 25 units along, so $\frac{3}{5}$ of the trip. **17.** (a) $\sqrt{15^2 + 8^2} = \sqrt{289} = 17$ km; slope $\frac{8}{15}$. (b) Midpoint $\left(\frac{3}{2}, 2\right)$; the service road has the negative reciprocal slope, $-\frac{15}{8}$. (c) $\frac{-10-5}{4-(-4)} = -\frac{15}{8}$, the same slope, so yes — parallel to the service road, and therefore perpendicular to the highway. **18.** (a) $\sqrt{36 + (y+2)^2} = 10 \Rightarrow (y+2)^2 = 64 \Rightarrow y = 6$ or $y = -10$: the points $(0, 6)$ and $(0, -10)$. (b) Slopes $-\frac{4}{3}$ and $\frac{4}{3}$: the same steepness, opposite directions — one segment falls to the right, the other rises. (c) No; $-\frac{4}{3} \times \frac{4}{3} = -\frac{16}{9} \neq -1$. Opposite signs are not enough. **19.** $(-1, 0)$. Distances: $(x+6)^2 + 9 = (x-2)^2 + 25 \Rightarrow 12x + 45 = -4x + 29 \Rightarrow x = -1$; check $\sqrt{25+9} = \sqrt{34} = \sqrt{9+25}$. Perpendicular bisector: midpoint $(-2, -1)$, $m_{AB} = -1$, so the bisector has slope 1 through $(-2, -1)$; going up 1 and right 1 reaches $(-1, 0)$, which is on the x-axis. **20.** $D(-1, -6)$ when AC is a diagonal (then BD is the other, and both are centred at $(-1, -\frac{5}{2})$); $D(-13, 0)$ when AB is a diagonal (centre $(-4, -1)$); $D(11, 2)$ when BC is a diagonal (centre $(2, -\frac{1}{2})$). In each case the test is the same: the two diagonals must share a midpoint.